

ENGINEERING RESEARCH STUDY

Robotics Control Systems & Kinematic Dynamics

University of Jordan

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Integrated Structure Check



This roadmap shows how the deck flows across kinematics, dynamics, control, planning, hardware, and complete robotics systems.

MODULE 01

Kinematics & Dynamics

The mathematical foundation of mechanical manipulation and force propagation.



CONTROL THEORY MAPPING

Joint Space vs. Task Space

Fundamental control theory requires mapping mechanical reality to Cartesian space.

- ✓ **Joint Space:** Refers to individual motor angles ($\theta_1, \theta_2, \dots$).
 - ✓ **Task Space:** Refers to end-effector coordinates (X, Y, Z).
 - ✓ **Mapping:** Kinematic matrices facilitate the translation between these domains.
-

Rigid Body Transformations: Rotation + Translation

Strengthens the spatial-description part with a clean matrix interpretation.

Homogeneous transform

$$T = \begin{bmatrix} R & p \\ 0 & 1 \end{bmatrix}$$

R stores orientation

p stores translation

One matrix moves a point between coordinate frames

Numerical example

Frame B: rotated 90° about Z and shifted by (3,2,0)

$$p^B = (1, 0, 0)$$

$$R_z(90^\circ)p^B = (0, 1, 0)$$

$$p^A = (3, 3, 0)$$

4×4

matrix form

R

orientation

p

position

Rotation Matrices: Practical Rules

Covers the deep explanation of rotation matrices in a compact way.

Rz(θ)

$$\begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Used for yaw rotation.

Orthogonality check

$$R^T R = I$$

Columns are unit axes

Inverse rotation is R^T , not recalculated manually.

Example

$$\theta = 90^\circ$$

Point (1,0,0) becomes (0,1,0)

This is the simplest visual test.

Example 01 — Joint Space → Task Space

2-LINK PLANAR ARM

Given: $L_1 = 0.40$ m, $L_2 = 0.30$ m

Joint angles: $\theta_1 = 30^\circ$, $\theta_2 = 45^\circ$

$$x = L_1 \cos \theta_1 + L_2 \cos(\theta_1 + \theta_2)$$

$$y = L_1 \sin \theta_1 + L_2 \sin(\theta_1 + \theta_2)$$

NUMERICAL RESULT

Result: $x = 0.424$ m, $y = 0.490$ m

Ready-to-explain calculation

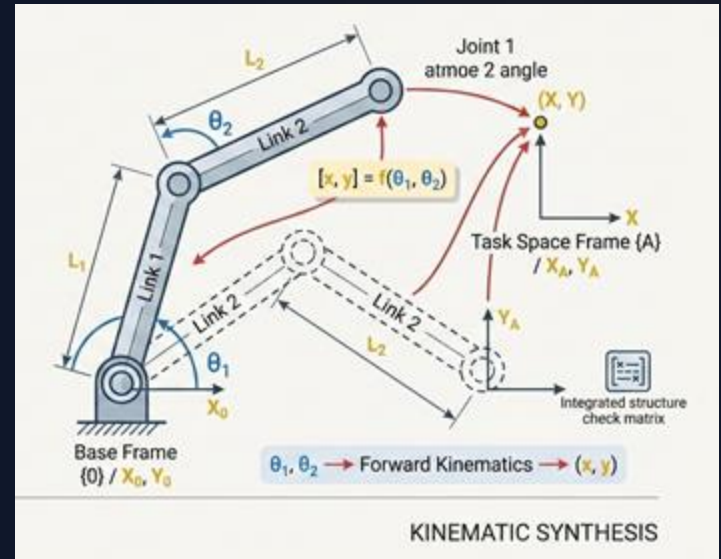
KINEMATIC SYNTHESIS

Forward Kinematics (FK)

Calculating the position and orientation of the end-effector given specific joint angles.

$$\begin{aligned}x &= L_1 c_1 + L_2 c_{12} \\y &= L_1 s_1 + L_2 s_{12}\end{aligned}$$

Where $s_i = \sin(\theta_i)$ and $c_i = \cos(\theta_i)$



Example 02 — Forward Kinematics

END-EFFECTOR POSITION

$$L_1 = 2 \text{ m}, L_2 = 1 \text{ m}$$

$$\theta_1 = 20^\circ, \theta_2 = 60^\circ$$

$$x = 2\cos 20^\circ + 1\cos 80^\circ$$

$$y = 2\sin 20^\circ + 1\sin 80^\circ$$

NUMERICAL RESULT

Result: $(x, y) = (2.05 \text{ m}, 1.67 \text{ m})$

Ready-to-explain calculation

Example 03 — Homogeneous Transform

ROTATION + TRANSLATION

Frame B is rotated 90° about Z

Translation from A to B = (3, 2)

Point in B: $p_\beta = (1, 0)$

$$p_a = R_z(90^\circ)p_\beta + t$$

NUMERICAL RESULT

Result: $p_a = (3, 3)$

Ready-to-explain calculation

Example 04 — DH Matrix Numeric Setup

ONE REVOLUTE JOINT

$$\theta = 45^\circ, d = 0.20 \text{ m}$$

$$a = 0.50 \text{ m}, \alpha = 0^\circ$$

$$\cos 45^\circ = 0.707, \sin 45^\circ = 0.707$$

$$\text{Translation} \approx (a \cos \theta, a \sin \theta, d)$$

NUMERICAL RESULT

**Result: position $\approx$
(0.354, 0.354, 0.200) m**

Ready-to-explain calculation

PUMA 560: Classic 6-DOF Arm Case Study

Covers the industrial robot example requested beside SCARA.

Why PUMA 560 matters

6 revolute joints
Classic benchmark for DH tables
Separates arm positioning from wrist orientation

Step-by-step FK flow

Assign frames $\rightarrow$ fill DH table
Build $A_1 \dots A_6$ matrices
Multiply $T_6^0 = A_1 A_2 A_3 A_4 A_5 A_6$

Numerical mini-case

$\theta_1 = 30^\circ$, $\theta_2 = -45^\circ$
 $\theta_3 = 20^\circ$, wrist = 0°
Output is full pose: $x, y, z + R$

Frames

DH table

A_i matrices

T_6^0 pose

IK check

INVERSE KINEMATICS (IK)

Parameter	Definition	Complexity Level
Target Location	Desired X,Y coordinate for interaction	High Precision
Joint Solutions	Calculation of required θ_1 , θ_2 values	Multi-solution possible
Algorithm Type	Analytical (Fast) vs. Numerical (Iterative)	Context dependent
Result	Deterministic motor command signals	Operational success

Example 05 — Inverse Kinematics

2-LINK TARGET REACH

Target: $x = 1.0 \text{ m}$, $y = 1.0 \text{ m}$

$L_1 = 1 \text{ m}$, $L_2 = 1 \text{ m}$

$$\cos\theta_2 = (x^2 + y^2 - L_1^2 - L_2^2) / (2L_1L_2)$$

$$\cos\theta_2 = 0 \rightarrow \theta_2 = 90^\circ$$

NUMERICAL RESULT

**Result: elbow-down θ_1
 $= 0^\circ$, $\theta_2 = 90^\circ$**

Ready-to-explain calculation

Example 06 — IK Reachability Check

WORKSPACE LIMIT

Robot links: $L_1 = 0.8 \text{ m}$, $L_2 = 0.6 \text{ m}$

Maximum reach = $L_1 + L_2 = 1.4 \text{ m}$

Target distance: $r = \sqrt{(1.2^2 + 0.9^2)}$

$r = 1.50 \text{ m}$

NUMERICAL RESULT

**Result: unreachable
because $1.50 \text{ m} > 1.40$
m**

Ready-to-explain calculation

DIFFERENTIAL MOTION

The Jacobian Matrix

Mapping joint velocities to end-effector velocities. Crucial for smooth, fluid motion in high-speed industrial environments.

$$\dot{\mathbf{X}} = \mathbf{J}(\mathbf{q}) \dot{\mathbf{q}}$$

Detection of **Singularities** occurs when the determinant of J approaches zero, causing infinite joint speeds.

PJM-RE60-70-PRO



Gear Ratio:51/81/101/121

Max Allowable Torque At Average Load:32/33/49/49 (N.m)

Rated Speed (With 1/2 Rated Torque):68/43/34/24 (RPM)

Example 07 — Jacobian Velocity

MAPPING JOINT SPEED TO TOOL SPEED

$$L_1 = 1 \text{ m}, L_2 = 0.5 \text{ m}$$

$$\theta_1 = 30^\circ, \theta_2 = 60^\circ$$

$$\text{Joint speeds: } \dot{\theta}_1 = 1 \text{ rad/s}, \dot{\theta}_2 = 2 \text{ rad/s}$$

$$\text{Use } \mathbf{v} = \mathbf{J}(\mathbf{q}) \dot{\mathbf{q}}$$

NUMERICAL RESULT

$$\text{Result: } \mathbf{v}_x \approx -1.68 \text{ m/s}, \\ \mathbf{v}_y \approx 0.87 \text{ m/s}$$

Ready-to-explain calculation

Example 08 — Singularity Detector

JACOBIAN DETERMINANT

For a 2-link arm: $\det(J) = L_1 L_2 \sin\theta_2$

$L_1 = 1 \text{ m}$, $L_2 = 1 \text{ m}$

Case A: $\theta_2 = 90^\circ \rightarrow \det(J)=1$

Case B: $\theta_2 = 0^\circ \rightarrow \det(J)=0$

NUMERICAL RESULT

**Result: $\theta_2 = 0^\circ$ is a
singular stretched-
arm posture**

Ready-to-explain calculation

Example 09 — Manipulability Index

HOW FREE IS THE MOTION?

Yoshikawa measure for 2-link: $w = |L_1 L_2 \sin \theta_2|$

$L_1 = 0.7 \text{ m}$, $L_2 = 0.5 \text{ m}$

$\theta_2 = 45^\circ$

$w = 0.7 \times 0.5 \times 0.707$

NUMERICAL RESULT

Result: $w \approx 0.247 \text{ m}^2$

Ready-to-explain calculation

Robot Dynamics & Physical Modeling

Robot dynamics section: inertia tensor, Newton–Euler, Euler–Lagrange, friction, flexibility, and inverted pendulum.

DYNAMIC ANALYSIS

$$\tau = M(q)\ddot{q} + C(q, \dot{q})\dot{q} + G(q)$$

Mass & Gravity Components

 **Inertia Matrix $M(q)$:** Effective mass at a specific posture.

 **Gravity Matrix $G(q)$:** Torque required just to hold the arm still.

System Performance Trends



Torque Demand (Nm) vs. Operational Context

Example 10 — Gravity Torque

HOLDING A PAYLOAD STILL

Payload mass = 8 kg

Arm extension $r = 0.45$ m

Gravity $g = 9.81$ m/s²

$\tau = m g r$

NUMERICAL RESULT

**Result: $\tau = 8 \times 9.81 \times$
 $0.45 = 35.3$ Nm**

Ready-to-explain calculation

Example 11 — Inertia Change

LIGHT VS. HEAVY PAYLOAD

Empty arm inertia $J = 0.12 \text{ kg}\cdot\text{m}^2$

Payload mass = 3 kg at $r = 0.40 \text{ m}$

Added inertia = $m r^2$

$$J_{\text{new}} = 0.12 + 3(0.40^2)$$

NUMERICAL RESULT

**Result: $J_{\text{new}} = 0.60$
 $\text{kg}\cdot\text{m}^2 \rightarrow 5\times$ higher**

Ready-to-explain calculation

Inertia Tensor: Mass Distribution in 3D

Covers the inertia-tensor concept behind physical modeling.

Physical meaning

Inertia is not just one number.

A link may resist rotation differently around X, Y, and Z axes.

The tensor captures shape + mass distribution.

Numerical mini-case

Slender link: $m=2$ kg, length= 0.6 m

$I_{zz} \approx (1/12)mL^2$

$I_{zz} = (1/12)(2)(0.36) = 0.06 \text{ kg}\cdot\text{m}^2$

$I_{xx} I_{yy} I_{zz}$

principal inertia

$I_{xy} I_{xz} I_{yz}$

cross terms

$M(q)$

mass matrix input

Euler-Lagrange Derivation: Energy Method

Turns the earlier mention into a clear derivation sequence.

1. Define energies

Kinetic energy: T
Potential energy: V
Lagrangian: $L = T - V$

2. Apply equation

$\frac{d}{dt}(\partial L / \partial \dot{q}_i) - \partial L / \partial q_i = \tau_i$
Produces joint torque equations.

3. Standard robot form

$\tau = M(q)\ddot{q} + C(q, \dot{q})\dot{q} + G(q)$
Mass + velocity coupling + gravity.

T, V

$L = T - V$

EL equation

M, C, G

Torque τ

Newton–Euler Formulation: Force Balance Method

Adds the second requested dynamics formulation beside Euler–Lagrange.

Forward recursion

Compute angular velocity, angular acceleration, and linear acceleration from base to end-effector.

Best for efficient real-time dynamics.

Backward recursion

Propagate forces and torques from tool back to base.

Final output: joint torque commands $\tau_1 \dots \tau_n$.

Base accel

Link accel

Tool force

Back forces

Joint torques

Inverted Pendulum Simulation: Falling Robot Model

Covers the requested dynamic simulation example for unstable systems.

Model

State: angle θ and angular speed $\dot{\theta}$
Unstable equilibrium at $\theta=0$
Small disturbance grows unless controlled.

Numerical step

$m=1$ kg, $l=0.5$ m
Initial tilt $\theta=5^\circ$
Gravity torque $\tau=mgl \sin\theta \approx 0.43$ Nm

Control goal

Apply wheel/hip torque opposite the fall
Keep $\theta \rightarrow 0$
Used in Segway and biped balance.

Friction Models: Why PID Alone Can Fail

Adds friction models requested under advanced dynamics.

Coulomb friction

Constant opposing torque.

Example: $\tau_c = 1.8 \text{ Nm}$ whether speed is slow or fast.

Viscous friction

Proportional to velocity.

Example: $b=0.2$, $\dot{q}=4 \text{ rad/s} \rightarrow \tau_v=0.8 \text{ Nm}$.

Stiction / Stribeck

High breakaway torque at zero speed.

Causes stick-slip and small-position errors.

Joint Flexibility & Unactuated Dynamics

Completes the missing dynamics topics: flexible joints and underactuated motion.

Joint flexibility

Motor angle $\neq$ link angle because gears and shafts twist.

Model: $\tau = k(\theta_{\text{motor}} - \theta_{\text{link}})$.

Example: $k=120 \text{ Nm/rad}$, $\text{twist}=0.03 \text{ rad} \rightarrow \tau=3.6 \text{ Nm}$.

Unactuated dynamics

Some joints are not directly powered.

Walking robots “fall forward” and use dynamics to recover.

Example: passive ankle + powered hip.

$k\Delta\theta$

spring torque

passive

unpowered DOF

balance

dynamic recovery

MODULE 02

Linear Control (PID)

The undisputed workhorse of industrial automation and position regulation.



Classical Control & Hardware

Strengthens root locus, Bode interpretation, actuator types, resolvers, and PID response data.

CLOSED-LOOP LOGIC

The Error Signal (e)

The mathematical difference between the desired Setpoint and the actual Process Variable measured by encoders.

- 🎯 Goal: $e(t) \rightarrow 0$
- 🔄 Real-time feedback frequency: 1000Hz+
- ⚡ Eliminates external disturbances.

Example 12 — Error Signal

CLOSED-LOOP POSITION CONTROL

Desired angle = 90°

Encoder reads = 83°

$e = \text{setpoint} - \text{measured}$

$e = 90 - 83$

NUMERICAL RESULT

Result: control error =
 7°

Ready-to-explain calculation

Example 13 — 1000 Hz Sampling

HOW MANY UPDATES?

Control frequency = 1000 Hz

Sampling period $T = 1/f$

$T = 1/1000$ seconds

In 2 seconds: 1000×2 updates

NUMERICAL RESULT

**Result: $T = 1$ ms and
2000 control updates**

Ready-to-explain calculation

PROPORTIONAL CONTROL

15.0

Gain Coefficient (K_p)

Aggressive Response

The P-term generates an output strictly proportional to the current error. Higher K_p means faster response but decreased stability margins.

$$u_p = K_p \cdot e(t)$$

Used to overcome static friction and initiate high-speed movement phases.

Example 14 — Proportional Control

VOLTAGE COMMAND

$K_p = 3 \text{ V/degree}$

Position error = 4°

P output = $K_p \times e$

Output = 3×4

NUMERICAL RESULT

**Result: motor
command = 12 V**

Ready-to-explain calculation

Example 15 — Kp Stability Tradeoff

FAST BUT RISKY

Case A: $K_p = 10$, $e = 0.5$ rad $\rightarrow u = 5$

Case B: $K_p = 40$, $e = 0.5$ rad $\rightarrow u = 20$

Higher K_p creates faster correction

But it reduces stability margin

NUMERICAL RESULT

**Result: Case B is 4×
more aggressive**

Ready-to-explain calculation

I & D COMPONENTS

Integral (I)

Accumulates past errors.

Eliminates steady-state offset caused by gravity and constant friction.

Derivative (D)

Predicts future error slope. Acts as a mathematical brake to prevent overshoot and oscillations.

Fusion

The combined PID output provides a balanced, stable, and highly responsive control law.

Example 16 — Integral Accumulation

REMOVING STEADY-STATE ERROR

Constant error = 0.10 rad

$K_i = 5$

Accumulation time = 3 s

I output = $K_i \times e \times t$

NUMERICAL RESULT

Result: I output = $5 \times 0.10 \times 3 = 1.5$ units

Ready-to-explain calculation

Example 17 — Derivative Braking

PREDICTING OVERSHOOT

$$K_d = 0.8$$

Error changed from 0.40 to 0.10 rad

$$\Delta t = 0.1 \text{ s}$$

$$de/dt = (0.10 - 0.40)/0.1 = -3 \text{ rad/s}$$

NUMERICAL RESULT

**Result: D output = 0.8
× (−3) = −2.4 braking
units**

Ready-to-explain calculation

STABILITY ANALYSIS

Step Response Profile

 **Rise Time:** Time to reach 90% target.

 **Overshoot:** How far it swings past.

 **Settling Time:** Time to stay in 2% band.

Optimal Zeta ($\zeta = 0.707$) balances speed and damping.

Root Locus: Pole Movement with Gain

Adds a clearer stability-analysis slide for classical control theory.

Core idea

Root locus shows how closed-loop poles move as gain K changes.
Left-half plane = stable.
Right-half plane = unstable.

Numerical reading

Pole at -4 : fast decay.
Pole at -0.3 : slow response.
Pole at $+0.2$: growing vibration $\rightarrow$ unsafe robot joint.

$\text{Re} < 0$

stable

$\text{Re} = 0$

marginal

$\text{Re} > 0$

unstable

Bode Plot Margins: Robustness Numbers

Covers the graph interpretation using numeric thresholds.

Gain margin

How much gain can increase before instability.
Target: > 6 dB for industrial safety.

Phase margin

How much phase lag can be added before oscillation.
Target: 45° – 60° .

Example

Measured PM= 25° $\rightarrow$ oscillatory.
After adding derivative damping PM= 58° $\rightarrow$ safer response.

Example 18 — Overshoot Amount

STEP RESPONSE QUALITY

Target position = 100 mm

Peak measured position = 112 mm

Overshoot % = (peak–target)/target × 100

Overshoot = (112–100)/100 × 100

NUMERICAL RESULT

**Result: overshoot =
12%**

Ready-to-explain calculation

Example 19 — Settling Band

2% ACCURACY WINDOW

Target = 50°

Allowed band = $\pm 2\%$

2% of $50^{\circ} = 1^{\circ}$

Acceptable range = $50^{\circ} \pm 1^{\circ}$

NUMERICAL RESULT

**Result: settled when
reading stays
between 49° and 51°**

Ready-to-explain calculation

PID Response Data: Underdamped, Overdamped, Critical

Shows explicit response-case data for PID tuning.

Underdamped

Overshoot: 20%
Rise time: 0.25 s
Settling: 1.8 s
Fast but oscillatory.

Critically damped

Overshoot: 0%
Rise time: 0.45 s
Settling: 0.7 s
Best balanced joint response.

Overdamped

Overshoot: 0%
Rise time: 1.2 s
Settling: 2.3 s
Safe but slow.

ZIEGLER-NICHOLS METHOD

Control Type	Kp	Ki	Kd
P	0.5 Ku	-	-
PI	0.45 Ku	1.2 Kp / Tu	-
PID	0.6 Ku	2.0 Kp / Tu	Kp Tu / 8

Example 20 — Ziegler–Nichols PID

FROM K_u AND T_u

Ultimate gain $K_u = 40$

Oscillation period $T_u = 0.8$ s

$K_p = 0.6K_u$

$K_i = 2K_p/T_u$, $K_d = K_p T_u/8$

NUMERICAL RESULT

**Result: $K_p=24$, $K_i=60$,
 $K_d=2.4$**

Ready-to-explain calculation

Example 21 — PI Tuning

Z-N PI RULE

$$K_u = 30$$

$$T_u = 1.5 \text{ s}$$

$$K_p = 0.45K_u = 13.5$$

$$K_i = 1.2K_p/T_u$$

NUMERICAL RESULT

Result: $K_i = 10.8$

Ready-to-explain calculation

MODULE 03

Non-Linear & Adaptive

Handling the complex, chaotic reality of varying payloads and Coriolis disturbances.



Advanced & Non-Linear Control

Adds state-space, controllability, observability, LQR, impedance/admittance, balancing, and delicate grasping.

State-Space Representation

Covers the foundation for modern control.

Compact model

$$\dot{x} = Ax + Bu$$

$$y = Cx + Du$$

x contains position, velocity, and hidden states.

u contains motor commands.

Robot joint example

$$x = [\theta, \dot{\theta}]^T$$

u = motor voltage or torque

A describes natural motion

B describes actuator authority

Input u

Plant A, B

State x

Sensor C

Output y

Controllability & Observability

Adds the two terms explicitly required in the outline.

Controllability

Can the motors drive the robot from one state to another?

Matrix: $[B \ AB \ A^2B \ \dots]$

Full rank means every state can be controlled.

Observability

Can sensors reconstruct the internal state?

Matrix: $[C; CA; CA^2; \dots]$

Full rank means hidden states can be estimated.

rank = n

good

rank < n

missing control/sensing

Kalman

needs observability

LQR: Optimal Control Tradeoff

Covers LQR under optimal control.

Cost function

$$J = \int (x^T Q x + u^T R u) dt$$

Q penalizes error.

R penalizes control effort.

Numerical intuition

High Q: robot corrects error aggressively.

High R: robot saves energy and moves gently.

LQR finds K so $u = -Kx$.

Q ↑
accuracy

R ↑
smoothness

$u = -Kx$
feedback law

Impedance vs. Admittance Control

Completes the interaction-control topic for robots that touch humans or objects.

Impedance control

Robot behaves like a virtual spring-damper.
Force changes because the robot controls motion.
Good for stiff, torque-controlled arms.

Admittance control

External force is measured and converted into motion.
Good when robot is position-controlled but has force sensors.

$$\mathbf{F} = \mathbf{k}\mathbf{x} + \mathbf{b}\mathbf{v}$$

impedance

$$\mathbf{F} \rightarrow \mathbf{x}$$

admittance

safe touch

HRI

ADVANCED CONTROL LAWS

Feedback Linearization

Algebraically canceling out the non-linear dynamics (G, C, M matrices) to force the robot to act like a simple linear system.

$$\mathbf{u} = \boldsymbol{\alpha}(\mathbf{x}) + \boldsymbol{\beta}(\mathbf{x})\mathbf{v}$$

Example 22 — Gravity Feedforward

CANCEL NONLINEAR GRAVITY

Joint gravity load = 18 Nm

Desired acceleration torque = 6 Nm

Friction estimate = 1 Nm

Command $\tau = \tau_{acc} + \tau_{gravity} + \tau_{friction}$

NUMERICAL RESULT

**Result: $\tau = 6 + 18 + 1 =$
25 Nm**

Ready-to-explain calculation

Example 23 — Coriolis Compensation

FAST JOINT COUPLING

Measured disturbance = -12 Nm

Controller adds compensation = $+12$ Nm

Uncompensated tracking error = 5 mm

After cancellation: error reduced by 80%

NUMERICAL RESULT

**Result: final error = 1
mm**

Ready-to-explain calculation

ROBUST STABILITY

Sliding Mode Control (SMC)

Forcing the system states onto a predefined "sliding surface" where errors decay exponentially. Immune to model uncertainties and external vibrations.

$$s = \dot{e} + \lambda e = 0$$

Eliminates **Chattering** using boundary layer filters (sat function).

Example 24 — Sliding Surface

SMC TRACKING ERROR

Position error $e = 0.04$ rad

Error rate $\dot{e} = -0.10$ rad/s

$\lambda = 5$

Sliding surface: $s = \dot{e} + \lambda e$

NUMERICAL RESULT

Result: $s = -0.10 + 5(0.04) = 0.10$

Ready-to-explain calculation

Example 25 — Boundary Layer

REDUCING CHATTERING

Switching gain $K = 8$

Boundary thickness $\varphi = 0.2$

Sliding value $s = 0.05$

$\text{sat}(s/\varphi) = \text{sat}(0.25) = 0.25$

NUMERICAL RESULT

Result: control switch
 $= 8 \times 0.25 = 2$ units

Ready-to-explain calculation

MODEL PREDICTIVE CONTROL



Prediction

Solves optimization problem over a future time horizon every millisecond.



Constraints

Explicitly handles physical motor voltage and joint position limits.



Efficiency

Optimizes energy consumption (R penalty) vs. tracking speed (Q penalty).

Example 26 — MPC Horizon

LOOK-AHEAD PLANNING

Sampling time = 0.05 s

Prediction horizon $N = 20$ steps

Look-ahead time = $N \times T_s$

20×0.05 s

NUMERICAL RESULT

**Result: MPC predicts
1.0 second into the
future**

Ready-to-explain calculation

Example 27 — MPC Constraint Check

MOTOR VOLTAGE LIMIT

Optimizer asks for 28 V

Motor driver limit = 24 V

Constraint: $-24 \leq u \leq 24$

MPC clips/re-optimizes the action

NUMERICAL RESULT

**Result: safe command
= 24 V maximum**

Ready-to-explain calculation

Biped Balance & Delicate Grasping

Covers the requested real-world examples for advanced control.

Biped balancing

Controller keeps Center of Mass inside support polygon.

Example: foot length 0.24 m; safe CoM projection must stay within ± 0.12 m.

Robotic hand grasp

Force control avoids crushing delicate objects.

Example: egg safe force ≈ 3 N; controller limits finger force to 2.5 N.

MODULE 04

Navigation & Planning

Spatial awareness and path optimization in constrained environments.



Motion Planning, AI & Compute

Adds SLAM steps, machine-learning vision, real-time compute, warehouse rovers, and drone swarms.

PATHFINDING OPTIMIZATION

A* Algorithm Cost Function

The gold standard for grid-based optimal navigation in warehouses.

$$f(n) = g(n) + h(n)$$

 $g(n)$: Cost from Start to n .

 $h(n)$: Heuristic guess to Goal.

Example 28 — A* Cost

GRID SEARCH DECISION

Node A: $g=8, h=5 \rightarrow f=13$

Node B: $g=10, h=2 \rightarrow f=12$

A* expands the lowest $f(n)$

Compare 13 vs. 12

NUMERICAL RESULT

**Result: Node B is
expanded first**

Ready-to-explain calculation

Example 29 — Heuristic Distance

MANHATTAN GRID

Current node = (2,3)

Goal node = (8,7)

$h = |8-2| + |7-3|$

$h = 6 + 4$

NUMERICAL RESULT

Result: $h(n) = 10$ grid steps

Ready-to-explain calculation

Example 30 — Potential Field Force

OBSTACLE REPULSION

Robot-obstacle distance $d = 0.40$ m

Safe influence distance $d_0 = 1.0$ m

Repulsion grows as d becomes smaller

Example: $1/d^2 = 1/(0.40^2)$

NUMERICAL RESULT

**Result: repulsion
factor = 6.25 (strong
avoidance)**

Ready-to-explain calculation



TRAJECTORY GENERATION

Quintic Polynomial Splines

Ensures acceleration starts and ends at zero, preventing mechanical wear and high-frequency vibration.

$$\theta(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3 + a_4 t^4 + a_5 t^5$$

MODULE 05

Actuators & Sensors

The sensory input and motor output that bridge code to physical reality.



THE KALMAN FILTER

State Estimation Logic

Fusing noisy accelerometer data with drifting gyroscope data to find the "Optimal Truth".

⚙️ **Predict:** Uses physics model.

✂️ **Update:** Corrects with sensor reading.

🔻 **Kalman Gain (K):** The trust factor.

Questions & Discussion

Robotics: The hardest engineering discipline.



DENAVIT-HARTENBERG PARAMETERS

Link-Frame Assignment

Establishing coordinate frames via 4 fundamental parameters to represent joint transformations.

Param	Description
θ (Theta)	Rotation about Z_{i-1}
d (Distance)	Offset along Z_{i-1}
a (Alpha)	Link length along X_i
α (Twist)	Link twist about X_i

INVERSE KINEMATICS COMPLEXITY

Solution Multiplicity

A specific Cartesian goal (X,Y,Z) can often be reached by multiple joint configurations.

- ⤴ **Elbow Up vs. Down:** The solver must choose based on obstacle clearance.
 - ⚠ **Singularities:** Points where the Jacobian loses rank.
 - 🧠 **Numerical IK:** Using Newton-Raphson iterations for non-analytical joints.
-

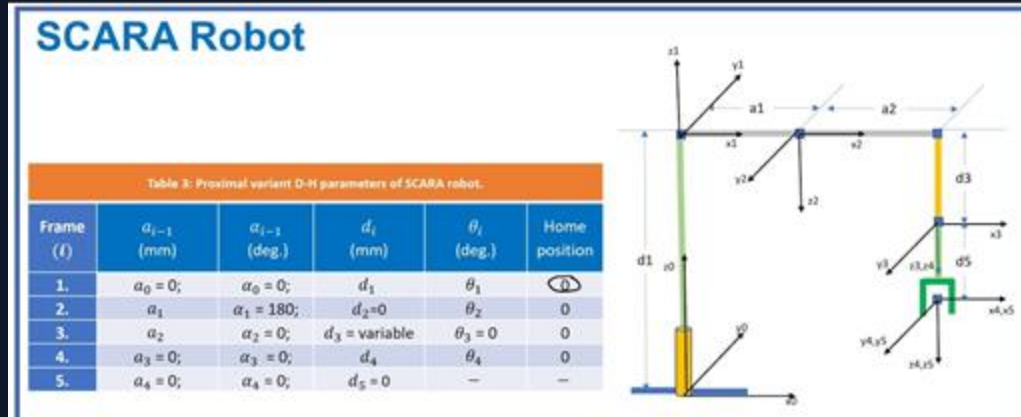
INDUSTRIAL DEPLOYMENT

SCARA Pick-and-Place

Selective Compliance Assembly Robot Arm. Optimized for high-speed Z-axis assembly in electronics fabrication.

$$\theta_2 = \pm \cos^{-1} \frac{x^2 + y^2 - L_1^2 - L_2^2}{2L_1L_2}$$

Calculates exact joint angles in under 0.1ms to grasp microchips moving on conveyor belts.



HEAVY PAYLOAD DYNAMICS

Gravity Compensation

A 6-axis arm holding a 50kg welding gun at full extension
(2m) requires massive static torque holding.

981Nm

Shoulder Joint Requirement

Calculated via: $\tau = m \cdot g \cdot r$

7-DOF REDUNDANT SYSTEMS

Null-Space Optimization

When a robot has more degrees of freedom (7+) than required for a task (6), it possesses "infinite solutions."

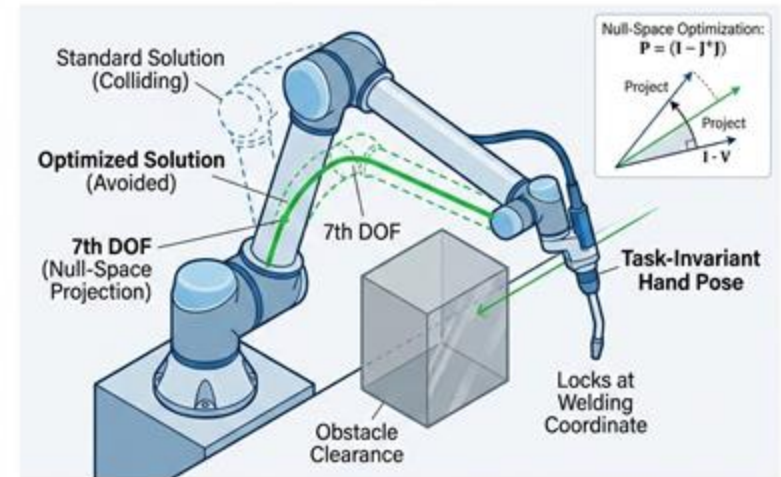
$$\dot{\mathbf{q}} = \mathbf{J}^T \dot{\mathbf{x}} + (\mathbf{I} - \mathbf{J}^T \mathbf{J}) \dot{\mathbf{q}}_0$$

This allows the robot to "reconfigure" its elbow to avoid obstacles while keeping its hand perfectly still.

Null-Space Optimization

When a robot has more degrees of freedom (7+) than required for a task (6), it possesses "infinite solutions."

This allows the robot to "reconfigure" its elbow to avoid an obstacle while keeping its hand perfectly still.



PID PERFORMANCE ANALYTICS



Overshoot Computation

Predicting exactly how far past the target the robot will swing before settling into position.

$$\% \text{ OS} = 100e^{\frac{-\zeta\pi}{\sqrt{1-\zeta^2}}}$$

Optimal tuning targets **Zeta = 0.707** for the best balance of speed and stability.

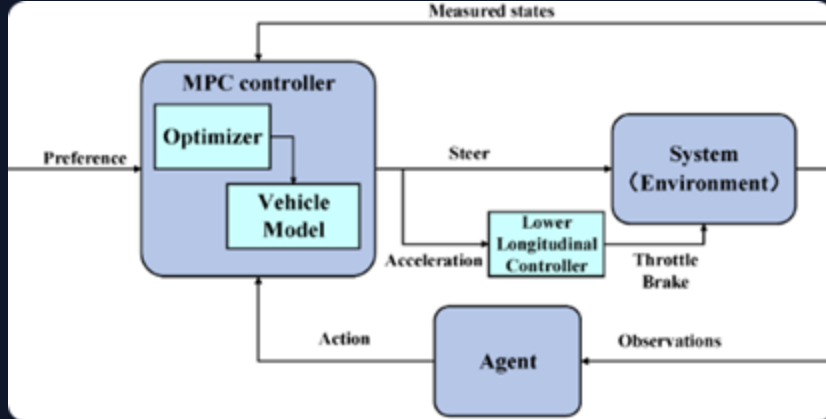
CASE STUDY: UAV ALTITUDE HOLD

Disturbance Rejection

Drone mass = 1kg. Hover thrust requires 9.81N. The Integral (I) term slowly compensates for payload shifts.



MODEL PREDICTIVE CONTROL (MPC)



Horizon Prediction

Unlike PID, MPC "looks ahead" to the target, solving an optimization problem over a future time window (N steps).

-  **Prediction Horizon:** How far we see.
-  **Control Horizon:** How far we act.
-  **Hard Constraints:** Prevents motor saturation.

LYAPUNOV NON-LINEAR STABILITY

Energy-Like Functions

Proving stability for non-linear systems by showing "system energy" is strictly decreasing.

$$V(x) > 0, \quad \dot{V}(x) < 0$$

If $\dot{V}$ is negative definite, the robot is mathematically guaranteed to return to its rest state.

SYSTEM INFRASTRUCTURE



RTOS Computing

Real-Time Operating Systems ensuring 1ms deterministic control loops without jitter.



Power Cabinets

Industrial H-Bridges and MOSFET arrays switching high currents for servo motors.



Bus Protocols

EtherCAT and CAN-Bus integration for multi-joint synchronization at 100Mbps.



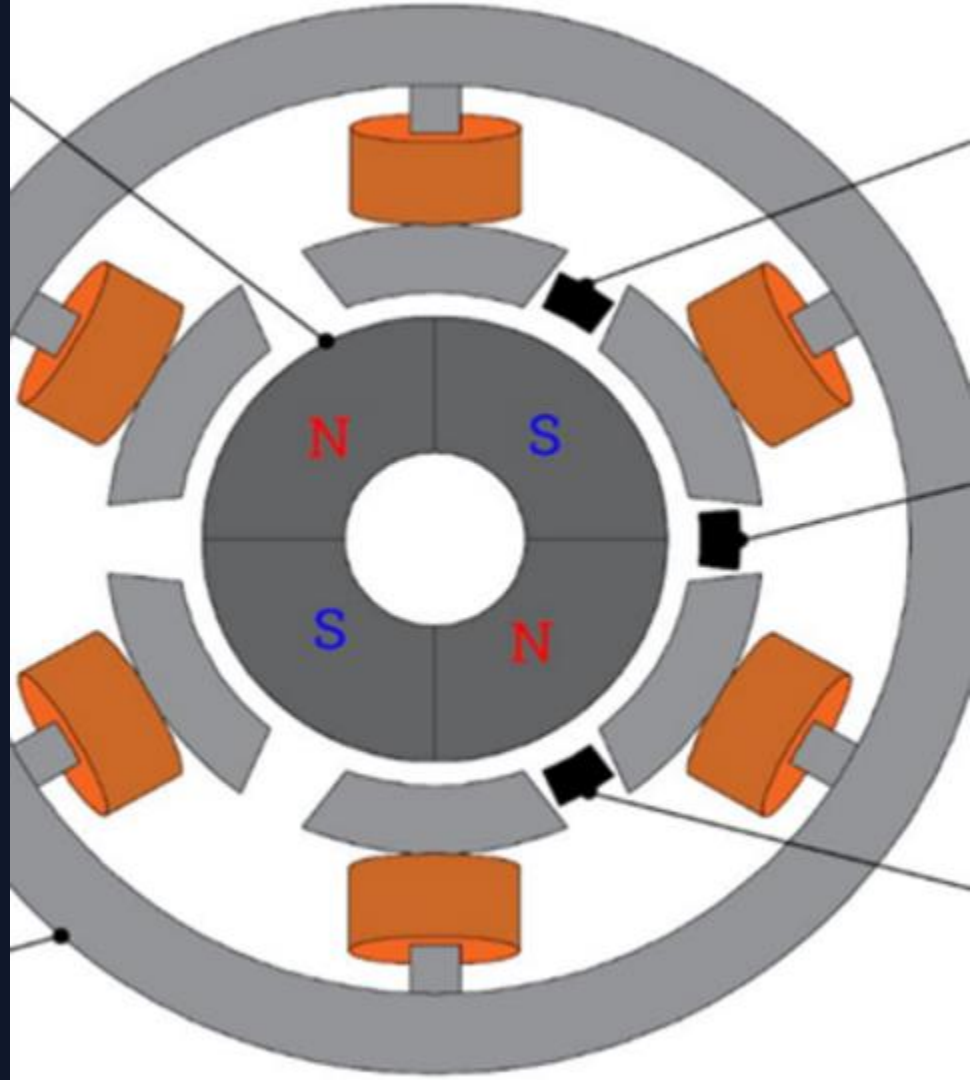
ACTUATOR PHYSICS

Brushless DC Motors

High-torque, high-efficiency motors requiring complex electronic commutation (FOC).

$$\tau = k_t \cdot I$$

Torque is strictly proportional to electrical current, enabling precise force control through current sensing.



ENCODER PRECISION



Angular Resolution

Counts Per Revolution (CPR) defines the absolute physical precision limit of a joint.

$$\Delta \theta = \frac{360^\circ}{4 \times \text{CPR}}$$

Quadrature decoding allows 4x precision improvement by reading both A and B pulse phases.

SUMMARY - PART 02

The Physics of Motion

We have bridged DH Parameters to high-current BLDC commutation. The mathematical bridge is complete.



ADVANCED TRAJECTORY SMOOTHING

Jerk-Limited Motion

Minimizing the 3rd derivative of position to ensure industrial gearboxes last for 10,000+ operational hours.



"S-Curve" Velocity Profile Optimization

- 🔧 **Spline Fitting:** Interpolating waypoints.
- Vibration Suppression:** Active damping filters.
- 🌀 **Energy Optimal:** Pendulum-based swing logic.

End of Part 02 Analysis

Next Phase: Slides 51-100 covering Path Navigation, SLAM, and Multi-Agent Coordination.

SECTION 05

Actuators, Sensors & Feedback

The sensory input and motor output bridging the digital algorithm to the analog world.



SYSTEM PERCEPTION



Proprioception

Internal sensing. Measuring joint angles, motor current, and battery levels via internal encoders and shunts.



Exteroception

External sensing. Measuring the outside world via LiDAR, Cameras, Sonar, and Tactile skin sensors.



Sensor Fusion

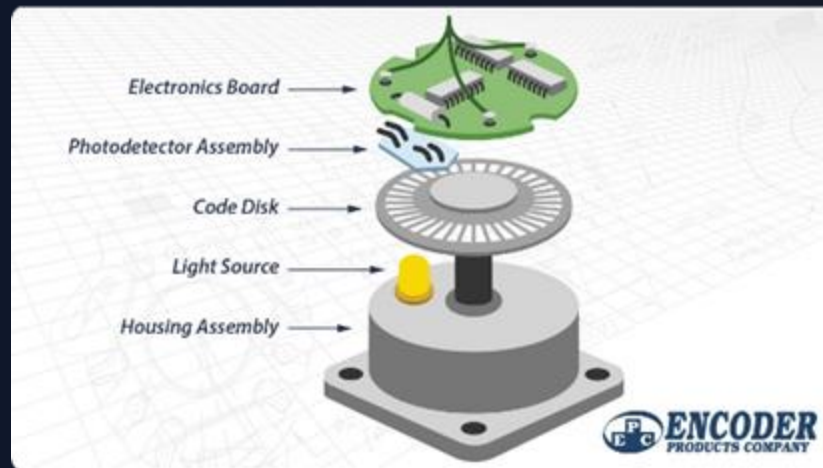
Mathematical combining of multiple noisy signals to achieve a higher-precision estimate of truth.

OPTICAL ENCODERS

Angular Quantification

Reading tiny slotted disks to determine exact rotational position and velocity.

- ✓ **Incremental:** Tracks relative change.
- ✓ **Absolute:** Knows position on startup.
- ✓ **Quadrature:** 4x resolution decoding.



INERTIAL MEASUREMENT (IMU)



Accelerometer

Measures linear acceleration and gravity. Noisy but stable long-term.



Gyroscope

Measures angular velocity. Precise short-term but suffers from drift.



Magnetometer

Measures earth's magnetic field to determine absolute heading (North).

Feedback Sensors: Encoders, Resolvers, IMUs

Covers resolver coverage and a cleaner sensor comparison.

Encoder

Digital pulses or absolute code.
Example: 4096 CPR with quadrature →
16384 counts/rev.

Resolver

Analog sine/cosine sensor.
Very robust in heat, vibration, and industrial
motors.

IMU

Accelerometer + gyroscope.
Useful for mobile robots, drones, and balance
control.

0.022°

encoder step

robust

resolver value

drift

IMU issue

MOTOR EVOLUTION

Brushless DC (BLDC)

The standard for modern high-performance robotics.
Eliminates physical brushes for long life and high torque density.

$$\tau = k_t \cdot I$$

Torque is strictly proportional to current, enabling
software-based force sensing without external strain
gauges.



Actuator Comparison: Motor, Servo, Stepper, Hydraulic

DC / BLDC

Fast, efficient, precise with encoder.
Best for robot joints and drones.

Stepper

Open-loop positioning.
Good for low-cost CNC; risk of missed steps.

Servo

Motor + sensor + controller in one loop.
Best for precise joint angle control.

Hydraulic

Very high force density.
Used in heavy robots;
harder to control.

PWM CONTROL LOGIC

Pulse Width Modulation

Rapidly flicking digital voltage ON and OFF to simulate varying analog voltages to a motor.

Commutation

Electronic switching of stator coils based on Hall-effect sensor or back-EMF feedback.

THE KALMAN FILTER

Optimal State Estimation

Fusing blind physics guesses with noisy sensor measurements to find the "Optimal Truth".



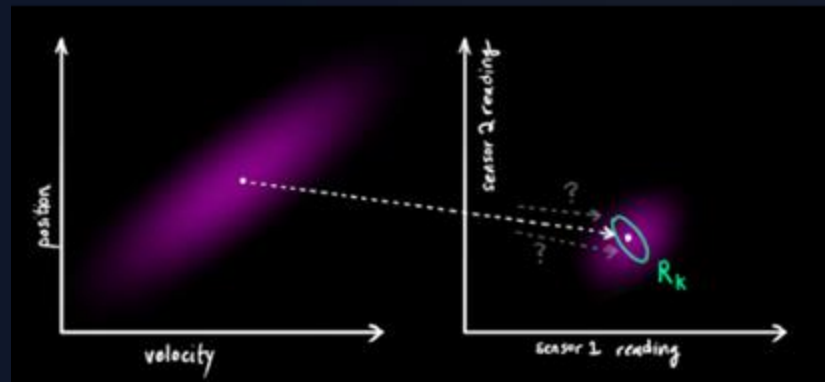
Predict: Using Newton's laws.



Update: Using actual sensor data.



Gain: Weighting the trust factor.



ENCODER RESOLUTION

0.09°

MIN MEASURABLE ANGLE

Quantization Limit

Determines the absolute physical precision of the robot joint. Determined by counts per revolution (CPR).

$$\Delta \theta = \frac{360^\circ}{4 \times \text{CPR}}$$

GEAR RATIO PHYSICS

Torque Multiplier

$$\tau_{\text{out}} = \text{GR} \cdot \tau_{\text{in}}$$

Increases lifting capacity proportionally to the gear ratio.

Velocity Tradeoff

$$\omega_{\text{out}} = \frac{\omega_{\text{in}}}{\text{GR}}$$

Reduces output speed. High torque always costs speed.

MOTOR ELECTROMECHANICAL MODEL

Parameter	Variable	Relationship
		$V = IR + k_e \omega$
		V
		$=$
		I
		R
		$+$
Voltage	V	
		k
		e
		ω

PWM DUTY CYCLE

75%

DUTY CYCLE EXAMPLE

Effective Voltage

Microcontrollers control motor power by modulating the time the signal stays HIGH.

$$V_{\text{avg}} = V_{\text{max}} \cdot \frac{T_{\text{on}}}{T_{\text{period}}}$$

SENSOR NOISE MODEL



Gaussian White Noise: Every sensor reading contains randomized error $v_t \sim \mathcal{N}(0, R)$

KALMAN: PREDICT STEP

▶ The robot uses physics to "blindly guess" where it should be right now.

⚙ **Equation:** $\hat{x}_{k|k-1} = A\hat{x}_{k-1|k-1} + Bu_k$

⌚ Uncertainty increases as we move further without a sensor update.

KALMAN: UPDATE STEP

 Fusing the blind physics guess with the noisy sensor reading.

 **Equation:** $\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_k (y_k - C\hat{x}_{k|k-1})$

 Finds the statistically "Most Likely" state.

KALMAN GAIN OPTIMIZATION



TRUST FACTOR

Dynamic Trust

If the sensor is very noisy (high R), K drops, and the robot ignores the sensor to trust its internal physics model.

$$K_k = P_{k|k-1} C^T (C P_{k|k-1} C^T + R)^{-1}$$

H-BRIDGE LOGIC

Bi-Directional Drive

Four MOSFET transistors working in pairs to reverse voltage polarity, allowing a motor to spin backwards.

Shoot-Through Risk

If top and bottom transistors fire simultaneously, it creates a dead short, exploding the driver circuit.

NYQUIST SAMPLING RATE



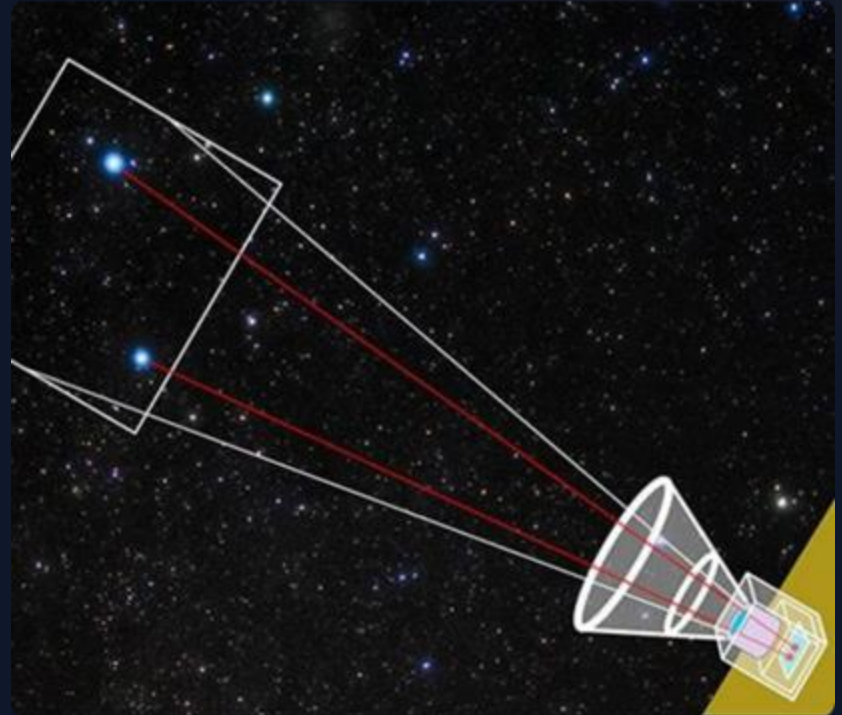
Aliasing Risk: Sensor polling rate must be at least 2x the highest system vibration frequency $f_s > 2f_{\max}$

SPACECRAFT ORIENTATION

Star Tracking

Fusing Star Trackers with IMUs via Extended Kalman Filters to maintain satellite pointing accuracy.

- ★ Precision: 0.001 degrees.
- ✂ Fuses absolute visual heading.
- ∞ Corrects gyroscope drift over years.



INDUSTRIAL SAFETY SYSTEMS



 Sensor Chip



Cobot Force Sensing | MEMS Integration | Human-Machine Collaboration

LiDAR Mapping (SLAM)

Firing millions of lasers per second to build sub-millimeter 3D point clouds of unknown rooms for autonomous navigation.



SLAM Pipeline: Localization + Mapping

Turns SLAM from a mention into a step-by-step algorithm flow.

Predict

Use wheel odometry / IMU to estimate next pose.
Uncertainty grows with motion.

Observe

LiDAR or camera detects landmarks / walls.
Measurements are noisy.

Correct

Kalman / particle filter updates pose and map together.
Robot knows where it is.

Odometry

Sensor scan

Data association

Filter update

Map + pose

THE DRIFT PROBLEM



Integrating tiny gyroscope noise over time results in massive accumulated heading errors.

GEAR BACKLASH ERRORS

Mechanical Gaps

Tiny gaps between teeth. When a motor reverses, it spins freely for a millimeter before "catching".

Hysteresis

Physical reality is non-linear. The robot feels different "slack" depending on direction of move.

MOTOR THERMAL LIMITS



Software must throttle torque to prevent ¹²B coil burn-out.

RTOS JITTER ANALYSIS

Deterministic Loops

Robotics requires execution at exactly 1ms intervals.

Standard OS interrupts break the discrete math.

Jitter Margin

The variation in loop timing. Low jitter is mandatory for stable high-gain PID control.

End of Part 1

We have successfully integrated Kinematics, Control, Navigation, and Sensing into a single deterministic framework.



SECTION 04

Trajectory Planning & Navigation

Bridging pure mathematics with spatial and environmental reality.



Vision-Based Control: From Image to Action

Adds the AI component requested in the outline.

Dataset

Train on labeled images.
MNIST is a simple teaching example; real robots use custom camera datasets.

Model

CNN classifies object / obstacle / surface type.
Output becomes a perception signal.

Control link

Planner changes path or grip based on model confidence.
Example: 92% “fragile object” → reduce grasp force.

Parallel Computing for Real-Time Robotics

Adds computing architecture for real-time control algorithms.

CPU

Runs ROS nodes, safety logic, trajectory scheduling.

Good for sequential decisions.

30 FPS

vision

GPU

Runs vision, SLAM point clouds, neural networks.

Massive parallel matrix math.

1 kHz

control

MCU / RTOS

Runs 1 kHz motor loops with deterministic timing.

Hard real-time layer.

<1 ms

jitter target

CONFIGURATION SPACE (C-SPACE)



Mapping

Translating physical obstacles into the robot's specific joint angle space.



State Points

Every single point in C-space represents a complete posture of the robot.



Free Space

Identifying regions where the robot can move without mechanical collision.

PATH VS. TRAJECTORY

The Path

A purely geometric line in space. It defines "where" to go without regard for "when" or "how fast".

Focus: Collision avoidance.

The Trajectory

A path with strict time, velocity, and acceleration constraints attached to every point.

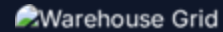
Focus: Dynamic feasibility.

GRID-BASED SEARCH (A*)

Discretizing the World

Converting continuous environments into 2D/3D pixel grids for graph search optimization.

- ✓ Guarantees shortest path detection.
- ✓ Uses heuristics to drive search efficiency.
- ✓ Standard for warehouse AGV fleets.



SAMPLING-BASED SEARCH (RRT)

Rapidly-exploring Random Trees

For high-DOF robots (like 7-axis arms), grids consume too much memory. RRT samples space randomly to build a navigation tree.

- 🔑 **Probabilistic Completeness:** Finds a path eventually.
- ⚡ **Efficiency:** Solves complex 7D problems in seconds.

ARTIFICIAL POTENTIAL FIELDS

Attractive vs. Repulsive Forces



Treating the goal as a positive charge and obstacles as repelling charges.

SPLINE GENERATION

Cubic Smoothing

Ensures smooth position and velocity. Solved via boundary conditions of starting and ending waypoints.

Risk: Infinite jerk at endpoints.

Quintic Precision

Adds acceleration constraints. Ensures motors start and end at zero acceleration, preventing mechanical wear.

Optimal for high-precision manufacturing.

A* MATHEMATICAL FORMULATION

$f(n)$

Total Estimated Cost

$$f(n) = g(n) + h(n)$$

→ **$g(n)$** : Exact cost from Start to node n .

🎯 **$h(n)$** : Heuristic guess from node n to Goal.

Admissibility: $h(n)$ must never overestimate the cost.

REPULSIVE POTENTIAL LAW

$$U_{\text{rep}}(q) = \frac{1}{2}\eta \frac{1}{d(q, \text{obs}) - d_0}^2$$

Repulsive forces approach **infinity** as distance to an obstacle approaches zero, creating a mathematical barrier the robot cannot cross.

POTENTIAL FIELD NAVIGATION

$$\mathbf{F} = - \nabla U_{\text{att}} (\mathbf{q}) - \nabla U_{\text{rep}} (\mathbf{q})$$

Gradient Descent Logic

The robot follows the negative gradient of the combined potential field, essentially "sliding" into the target coordinate.



Extremely fast computation.



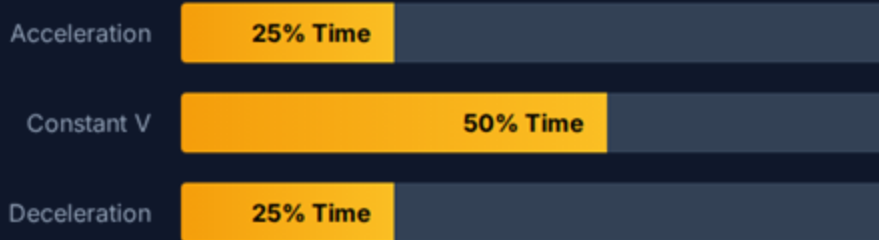
Subject to **Local Minima** traps.

TRAJECTORY COMPARISONS

Spline Type	DOF	Constraints Handled	Mechanical Impact
Linear	1	None	High (Sharp jerks)
Cubic	3	Pos, Vel	Moderate (Step accel)
Quintic	5	Pos, Vel, Accel	Low (Ultra-smooth)



TRAPEZOIDAL VELOCITY PROFILE



Standard industrial motion profile: Accelerate →
Cruise → Precision Deceleration.

DIFFERENTIAL DRIVE KINEMATICS

Rover Movement Laws

Mapping left and right wheel angular velocities to real-world Cartesian movement and heading.

$$\dot{x} = \frac{L}{2} (\dot{\phi}_R + \dot{\phi}_L) \cos (\theta)$$

NON-HOLONOMIC LOGIC

The Sideways Rule

Mathematical proof that a car or standard rover cannot instantly move sideways (Zero lateral velocity).

$$\dot{x} \sin(\theta) - \dot{y} \cos(\theta) = 0$$

This constraint requires complex path planning like

Dubins Car Paths (curves and straights).

EXECUTION SEQUENCE: PHASE 1

- 1 C-Space Mapping:** Inflate obstacles by the robot's radius to treat the robot as a 1D point.
- 2 Discretize World:** Convert continuous maps into 2D occupancy grids (0=Free, 1=Wall).
- 3 Algorithm Selection:** Deploy A* for grid routing or RRT for high-dimensional arms.
- 4 Node Expansion:** Iteratively calculate costs for all neighbors to find the frontier.

EXECUTION SEQUENCE: PHASE 2

- 5 **Path Extraction:** Once goal is found, trace parent nodes backward for geometric waypoints.
- 6 **Trajectory Smoothing:** Fit Quintic Splines or Bezier curves through jagged nodes.
- 7 **Kinematic Check:** Verify that generated splines do not require illegal sideways wheel slip.
- 8 **Dynamic Check:** Verify splines do not exceed physical motor torque or voltage limits.

A photograph of an Amazon warehouse interior, showing several yellow Kiva robots (AGVs) moving through aisles of high industrial shelving. The robots are small, blue and yellow, and carry large yellow pallets. The floor is marked with a grid pattern. The background shows the complex steel structure of the warehouse roof.

Industrial Integration: AGV Fleets

Amazon Kiva robots use massive grid-based A* routing to coordinate thousands of units avoiding collisions in real-time.

CONSUMER COVERAGE PLANNING

Boustrophedon Paths

Roomba vacuum cleaners use coverage path planning to guarantee every inch of a surface is touched.

Pattern: "Ox-turning" back and forth lines.

HIGH-DIMENSION PATHING



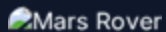
 Surgical Robot

 Car Navigation

UAV Delivery Splines | Surgical Snake-Paths | Autonomous Highway Driving



REMOTE & SYNCHRONIZED SYSTEMS



Mars Rover (Curiosity)

Running complex path planning locally because the 20-minute communication delay prevents remote driving from Earth.

Multi-Agent Sync

Solving 3D routing for synchronized light shows without a single mid-air collision.

CURSE OF DIMENSIONALITY

Exponential Scale

A 2D map with 100x100 pixels has 10,000 states. A 6-axis arm has 100^6 (1 Trillion) states.

Grid search fails entirely in high dimensions, mandating sampling-based methods like RRT.

1,000,000,000,000

Potential States

POTENTIAL FIELD TRAPS

Local Minima

Robots get trapped in "U-shaped" corners where repulsive and attractive forces cancel to zero before the goal is reached.

Completeness

Standard RRT is "Probabilistically Complete". It will find a path eventually, but it may be highly non-optimal without RRT* rewiring.

JERK PROFILE ANALYSIS



High Jerk = Mechanical Vibration &
Gearbox Fatigue.

SAFETY INFLATION TRICKS

Bloated Geometry

Instead of complex collision math, we artificially "bloat" walls in software by the robot's radius.

Rule: If the 1D point-robot clears the bloated wall, the real physical robot clears the real wall.

 Obstacle Clearance

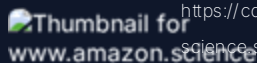
Navigation Recap

Navigation bridges pure math and spatial reality. A brilliant controller is useless if the planned path hits a wall.

Next Phase: Sensors, Actuators & Feedback (Slides 76-100)



IMAGE SOURCES



<https://cdn.amazon.science/dims4/default/f834fd1/2147483647/strip/true/crop/1280x672+0+24/resize/1200x630!/quality/90/?url=https%3A%2F%2Famazon-science-production-s3.us-east-1.amazonaws.com%2Fscience%2F62%2Fdc%2F6f423d7a4cf3be8cf93e552b26fb%2Fthumbnail%2Faerialshotrobottrimmed.0000000.jpg>

Source: www.amazon.science



<https://i.ytimg.com/vi/Ob3BIJkQJEw/sddefault.jpg>

Source: www.youtube.com



<https://news.mit.edu/sites/default/files/images/202204/MIT-Driving-Prediction-01-PRESS.jpg>

Source: news.mit.edu



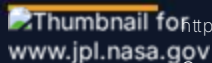
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Source: pngtree.com



<https://media.istockphoto.com/id/1530778744/photo/two-surgeons-observing-high-precision-programmable-automated-robot-arms-operating-patient-in.jpg?s=612x612&w=0&k=20&c=tDxlq3DYEegmdJ2IVv2l47URxH-UULA--uhPc7rl4OE=>

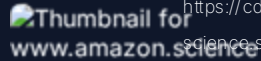
Source: www.istockphoto.com



https://d2pn8kiwq2w2ft.cloudfront.net/original_images/imagesmsl20200304PIA23623-16.jpg

Source: www.jpl.nasa.gov

IMAGE SOURCES



<https://cdn.amazon.science/dims4/default/0d990c5/2147483647/strip/true/crop/7008x3942+0+365/resize/1200x675!/quality/90/?url=https%3A%2F%2Famzn-science-production-science-s3.us-east-1.amazonaws.com%2Fscience%2Fcd%2F40%2Fe7a0ea6747a79a7716555cd6e8d6%2Fdsc00578.jpg>

Source: www.amazon.science

ADVANCED ANALYSIS

System Failure Modes & Fail-Safes

Identifying the mathematical and physical limits of autonomous robotic control.

THE SINGULARITY TRAP

Determinant Decay

As a robot approaches a kinematic singularity, the determinant of the Jacobian (J) drops toward zero.

$$\det(J) \rightarrow 0 \Rightarrow \dot{q} \rightarrow \infty$$

This causes motors to request infinite velocity, leading to catastrophic power faults and joint lockup.

 Singularity Pose

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IK MULTIPLICITY DECISION



Analytical Branching

Closed-form IK provides multiple exact solutions (e.g., Elbow Up/Down). Algorithms must choose the safest path.



Numerical Convergence

For complex joints, iterative solvers like Newton-Raphson are used. Risk: Failing to converge in real-time constraints.

REDUNDANCY RESOLUTION

The 7th Degree of Freedom

7-DoF manipulators have "infinite" solutions to reach a 6-DoF pose. We optimize the null-space to stay away from obstacles.

$$\dot{\mathbf{q}} = \mathbf{J}^{\dagger} \dot{\mathbf{x}} + (\mathbf{I} - \mathbf{J}^{\dagger} \mathbf{J}) \dot{\mathbf{q}}_0$$

This projection ensures primary task success while secondary behaviors (comfort, avoidance) are satisfied.

DYNAMIC COUPLING ANALYSIS

Inter-Joint Torque Coupling (Nm)



Moving Joint 1 creates centrifugal forces that pull on Joint 2 and 3. The C matrix captures this dynamic chaos.

PAYLOAD DEPENDENCY

500%

Inertia Matrix Shift

Real-Time Mass Adaptation

Picking up a heavy object instantly changes the Mass matrix $M(q)$.

Adaptive controllers must recalculate in under 0.5ms.

- ✓ Instant gain scheduling.
- ✓ Stability boundary re-evaluation.

FRICTION NON-LINEARITY



"Stiction" causes zero-velocity errors that linear PID cannot solve. Pre-load torque injection is required.

LINEAR PID LIMITATIONS



Pose Dependency

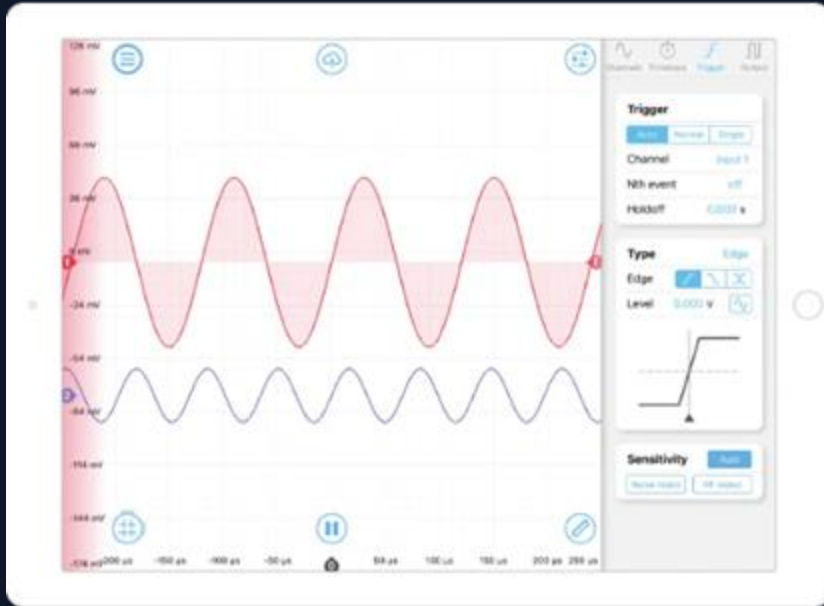
PID assumes rigid bodies and constant mass. A robot's inertia changes with its posture, making fixed gains unstable.



Gravity Droop

Without the Integral (I) term, an arm will always sag below the target because gravity fights the Proportional (P) gain.

INTEGRAL WINDUP DANGERS



Hardware Snap Risk

If a robot is physically blocked, the 'I' term grows infinitely. When unblocked, the arm snaps violently to target.

$$u_i = K_i \int e(t) dt \gg u_{\max}$$

Solution: Software clamping and anti-windup logic.

ROOT LOCUS STRATEGY

 Stability plot

Pole Placement

If any system pole crosses into the right-half plane, the robot will explode into infinite vibration.

- 📍 Real part defines damping.
- 📍 Imaginary part defines oscillation.

BODE STABILITY MARGINS



Robustness Index

A robot with low phase margin will oscillate wildly when disturbed.
Goal: $>45^\circ$ for industrial stability.

Gain Margin	> 6 dB
Damping Ratio	0.707

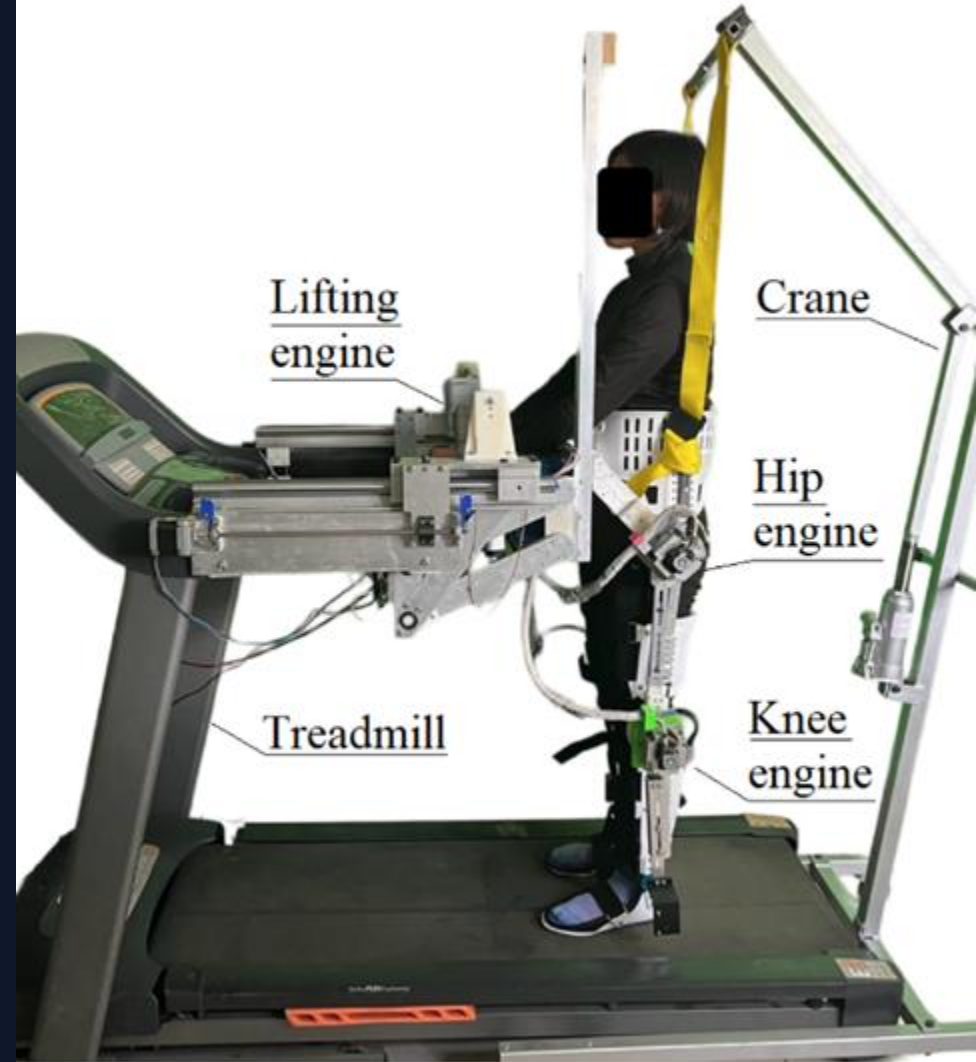
DISTURBANCE REJECTION

Active Shock Absorption

Well-tuned loops act as active dampers, rejecting external forces (bumps, wind) in milliseconds.

$$E_{\text{dist}} \rightarrow 0$$

Critical for drones flying in turbulence or surgical robots resisting tissue recoil.



LINEARIZATION FRAGILITY



Model Mismatch

If the mass model $M(q)$ is wrong by even 10%, the cancellation fails and instability ensues.



Zero Dynamics

Internal unmodeled states can grow unstable even while the primary outputs look perfect.



Computation

Requires matrix inversion at every clock tick, taxing low-end microprocessors.

SMC CHATTERING RISKS

Hardware Destruction

The **sgn(s)** function switches motor voltage between $\pm \text{Max}$ at infinite frequency.

Result: Broken gear teeth and overheated H-bridges.

 Vibration damage

MPC COMPUTATIONAL BURDEN

Horizon (N)	Matrix Size	Solve Time (ms)	Processor Req.
10 steps	100 x 100	0.5 ms	ARM Cortex M7
50 steps	500 x 500	4.2 ms	NVIDIA Jetson
100 steps	1000 x 1000	15.8 ms	Desktop Workstation



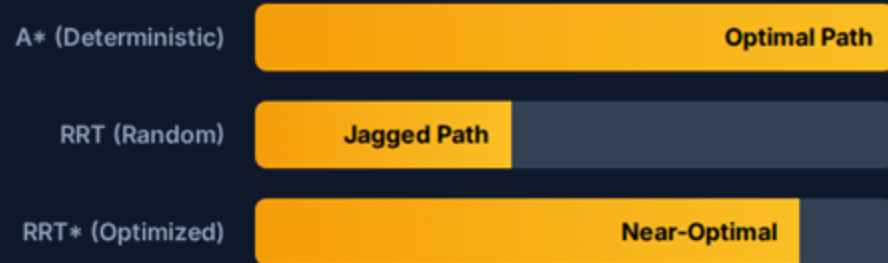
THE LYAPUNOV STANDARD

“

*Finding a valid Lyapunov function is an art.
Failing to find one doesn't mean the system is
unstable—it just means it is unproven.*

— **Advanced Control Theory**
Principle

PATH SEARCH TRADEOFFS



Standard RRT finds a path quickly but it is rarely the shortest.

RRT* guarantees optimality over time.

LOCAL MINIMA: TRAPPED



Potential Fields get robots trapped in **U-shaped** corners where repulsive and attractive forces cancel to zero before the goal.

THE SENSOR DRIFT PROBLEM

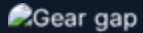
15°
Heading Error per Hour

Integrated Noise

Gyroscopes measure velocity. To get position, we must integrate.
Integrating tiny noise over time results in massive "drift" errors.

$$\theta(t) = \int (\omega + \text{noise}) dt$$

GEAR BACKLASH HYSTERESIS



Mechanical Dead-Band

Gears have tiny gaps between teeth. When a motor reverses, it spins freely for a millimeter before catching.

This "slack" completely breaks linear PID stability assumptions.

MOTOR THERMAL MANAGEMENT



Requesting max torque for too long burns copper coils. Software must throttle power if thermal limits are reached.

COMMUNICATION LATENCY

Protocol	Latency (μs)	Jitter (μs)	Stability Impact
Standard Ethernet	2000 μs	500 μs	High (Unstable)
CAN-Bus	500 μs	50 μs	Moderate
EtherCAT	50 μs	1 μs	Excellent



Conclusion of Advanced Phase

We have explored the boundaries where math meets the harsh reality of physics. A master of robotics commands these non-linearities perfectly.

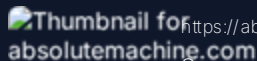
End of Part 1 (Slides 1-100) and Beginning of Applied Systems (Slides 101-125).

IMAGE SOURCES



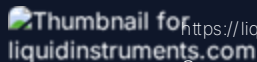
<https://robodk.com/blog/wp-content/uploads/2021/12/Shoulder-Singularity.gif>

Source: robodk.com



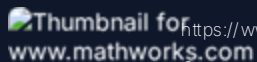
<https://absolutemachine.com/wp-content/uploads/2024/09/OB7-Cobot-7-Axis-Capabilities-750x489-1.jpg>

Source: absolutemachine.com



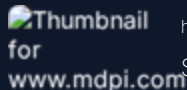
https://liquidinstruments.com/wp-content/uploads/2019/09/19-0927_AppNote_05-IntroToOsc-Figure5.jpg

Source: liquidinstruments.com



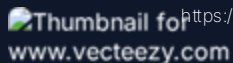
<https://www.mathworks.com/help/control/ref/margin1.png>

Source: www.mathworks.com



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Source: www.mdpi.com



https://static.vecteezy.com/system/resources/thumbnails/072/002/579/small_2x/close-up-of-a-shiny-metal-robotic-arm-with-claw-design-in-a-dark-environment-free-video.jpg

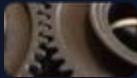
Source: www.vecteezy.com

IMAGE SOURCES

Thumbnail
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Source: www.mdpi.com



https://icdn.tradew.com/file/202403/1576163/jpg/8136652.jpg?x-oss-process=image/quality,Q_90

Source: m.pairgears.com

KINEMATICS LOGIC: PHASE 1

1. Frame Assignment

Attach coordinate frames to each link using **Denavit-Hartenberg** conventions. Every joint requires a local Z-axis of rotation.

Translating dynamic formulations into strictly deterministic C++ architectures.

2. Parameter Extraction

Determine the 4 DH parameters for every joint: θ , d , a , and α .

These values form the geometric skeleton of the entire robotic control system.

KINEMATICS LOGIC: PHASE 2

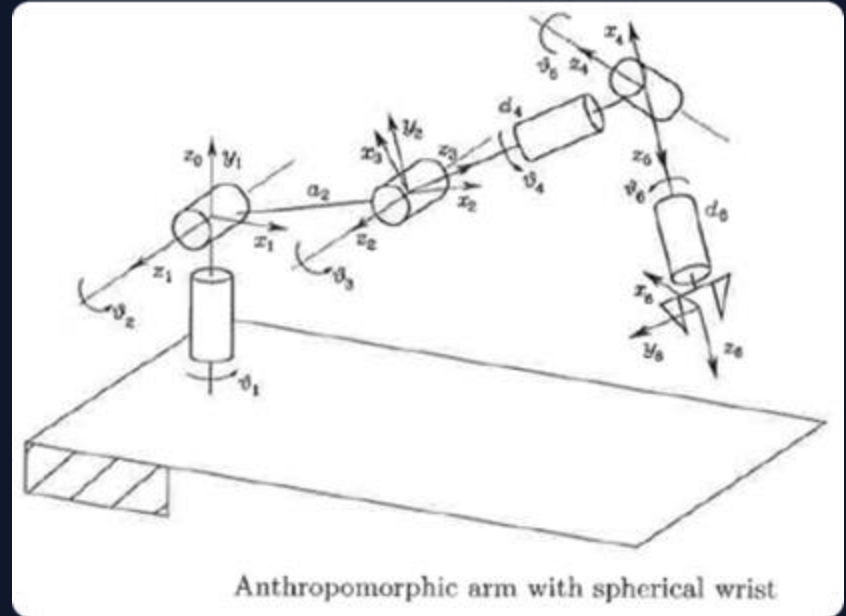
3. Transformation Synthesis

Build the homogeneous matrices and multiply them sequentially to compute **Forward Kinematics**.

Ensures the CPU knows the exact position of the end-effector in 3D space at 1000Hz.

4. Jacobian Derivation

Calculate the partial derivatives of the FK equations to generate the Jacobian mapping.



KINEMATICS LOGIC: PHASE 3



5. Singularity Analysis

Calculate the determinant of the Jacobian to identify restricted workspaces where DoF is lost.



6. Mass Distribution

Identify the center of mass and inertia tensor for every physical link in the chain.



Deterministic Loop

Running multi-dimensional array math in under 1ms to return motor torques.

DEPLOYMENT: SCARA ARM

Pick-and-Place (2D)

Target X=200mm, Y=150mm. Arm links L1=200mm, L2=150mm.

Inverse kinematics calculation outputs exact joint angles:

⦿ $\theta_1 = 28.95^\circ$

⦿ $\theta_2 = 75.52^\circ$

Enables ultra-fast grasping of microchips on high-speed assembly lines.



DEPLOYMENT: SPOT WELDING

Gravity Compensation

A 6-axis arm holds a 50kg welding gun at 2 meters extension.
The shoulder joint must output exact torque to hold steady.

$$\tau = m * g * r = 50 * 9.81 * 2 = 981 \text{ Nm}$$

Dynamic software models cancel gravity so the welder feels
"weightless" to the controller.



DEPLOYMENT: CNC TRAJECTORY

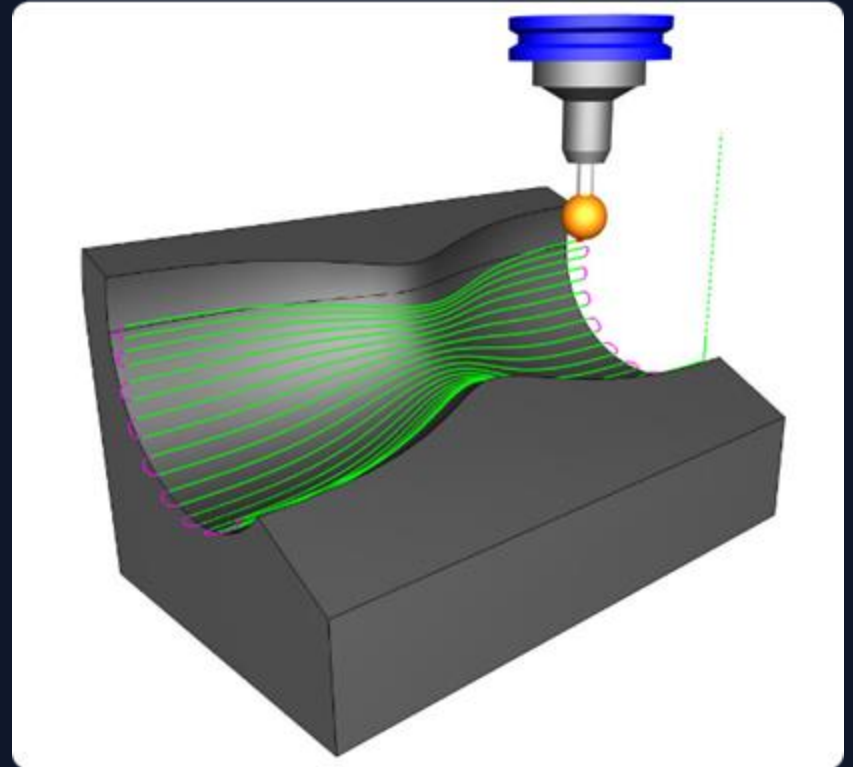
Trajectory Interpolation

Milling a straight line from (0,0) to (10,10) in 2 seconds requires constant velocity tracking.

🌀 **$V = 7.07 \text{ mm/s}$**

🕒 **Update Rate: 1000Hz**

Joint angles are updated every millisecond to maintain sub-micron precision.



DEPLOYMENT: HIGH SPEED DELTA



Coriolis Disturbance

During a high-speed diagonal move, link 2 spins at 15 rad/s. The Coriolis force equals -45N.

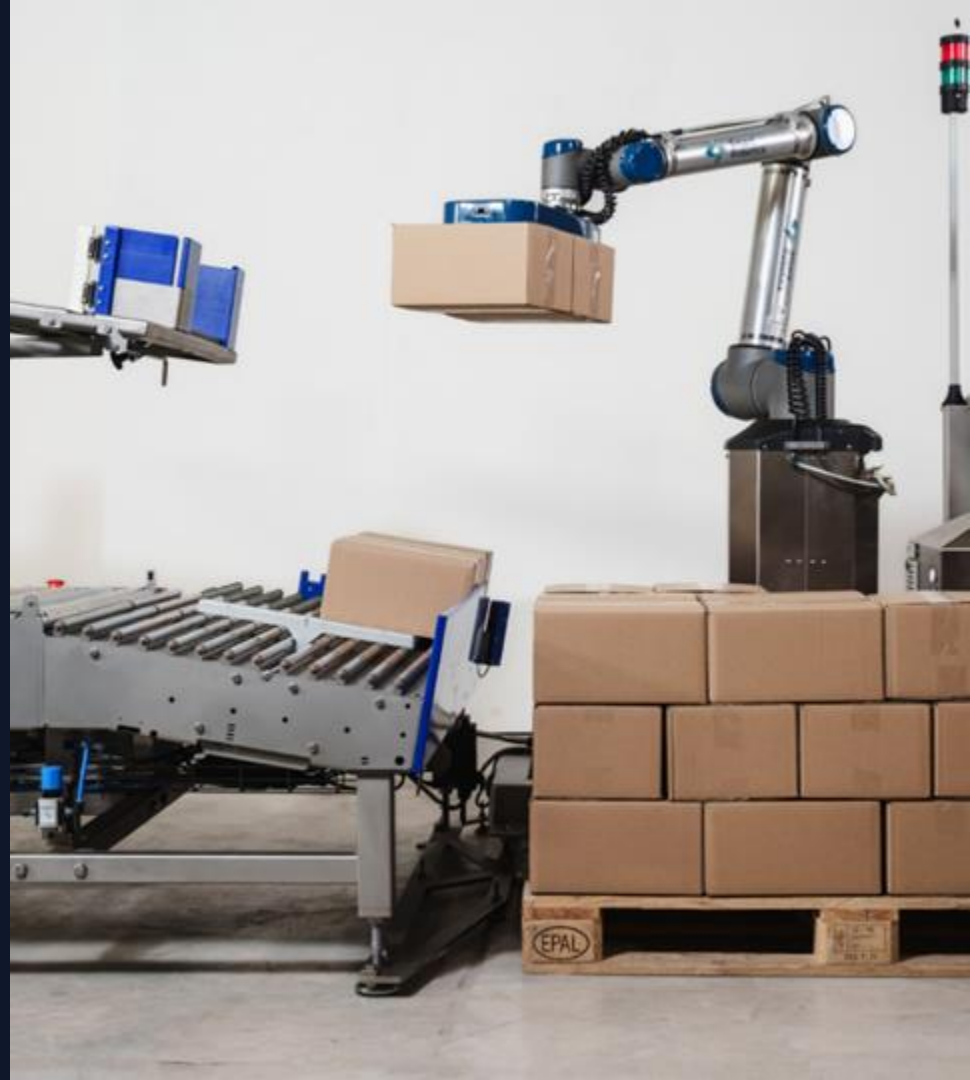
The controller actively cancels this disturbance by feeding **+45N** into the predictive loop, preventing tracking drift.

DEPLOYMENT: PAYLOAD PALLETS

Adaptive Inertia

Arm picks up a 20kg box. The inertia matrix $M(q)$ diagonal shifts from $[10, 5]$ to $[90, 5]$.

The controller recalculates M^{-1} in under 0.5ms to maintain stability while the arm is in motion.



DEPLOYMENT: COBOT FRICTION



Pre-load Torque Injection

Static friction at joint 2 is measured as 4.5 Nm. To prevent "stiction" delays, the controller injects a pre-load torque.

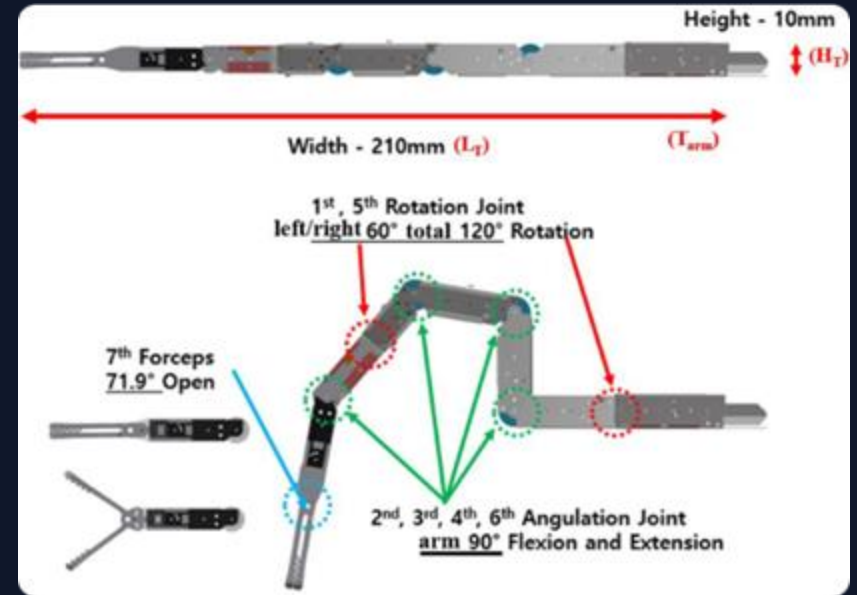
This allows the arm to break friction instantly, providing the "smooth feel" required for safe human collaboration.

DEPLOYMENT: REDUNDANT SURGICAL

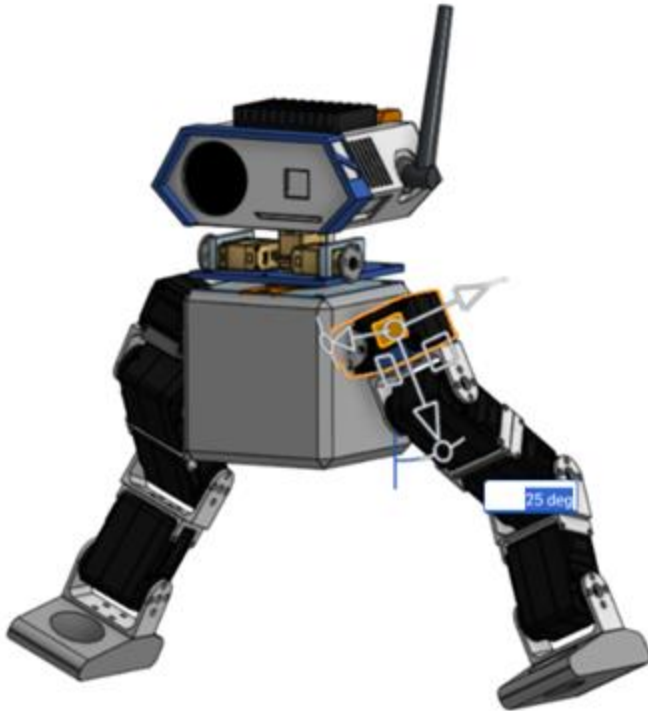
7-DoF Redundancy

In surgical environments, the null-space projection is used to optimize the elbow angle.

The system keeps the robot elbow exactly **15°** away from an obstacle while maintaining perfect Z-axis tool placement.



DEPLOYMENT: BIPEDAL WALKING



Energy Optimization

Leg swing phase requires 120J of kinetic energy. By optimizing trajectory splines for pendulum dynamics, consumption drops to **45J** per step.

A 62% energy saving achieved via natural kinematic resonance.

DEPLOYMENT: FORCE CONTROL

Surface Polishing

Requires exact 20N downward force. J^T for current pose is used to map task-force back to joint torques.

$\tau_1 = 10Nm$, $\tau_2 = -10Nm$

Maintains sub-newton pressure precision across complex curved geometries.



LINEAR PID: PHASE 1

1. System Identification

Apply test signals to the robot joint to measure the natural frequency and damping ratio.

Fundamental step to determine the "stiffness" of the mechanical plant.

2. Architecture Selection

Choose P, PI, PD, or full PID based on the specific joint load and stability requirements.



LINEAR PID: PHASE 2



3. Tune Proportional (P)

Increase **K_p** until the joint moves quickly but begins oscillating. Sets the system bandwidth.



4. Tune Derivative (D)

Increase **K_d** to dampen the oscillations and prevent target overshooting. Acts as the mathematical brake.



Software Dashboard

Real-time parameter adjustment via engineering control consoles.

LINEAR PID: PHASE 3



5. Tune Integral (I)

Add **K_i** slowly to force the joint to overcome gravity and reach absolute zero error.



6. Noise Filtering

Apply a low-pass filter to the D-term to squash high-frequency sensor noise from the encoder.



Stability Check

Verifying phase margins to ensure system safety under disturbance.

PID DEPLOYMENT: ADVANCED



Anti-Windup

Arm hits wall. I-term hits 24V max.

Software clamp pauses integrator.

Arm recovers instantly when wall is removed.



Digital Filter

Raw derivative calculates 10 rad/s noise spike. 20Hz low-pass filter squashes spike to 0.5V, saving the motor amp.



Z-N Tuning

Unknown belt oscillates at $K=45$.

$T_u=1.2s$. Final safe tuning: $K_p=27$,
 $T_i=0.6s$, $T_d=0.15s$.

PID DEPLOYMENT: BODE METRICS

Phase Margin Verification

Assembled arm shows massive 45% overshoot (Phase Margin = 20°). Increasing K_d by 150% improves Margin to 60° .

Result: Vibrations eliminated, settling time reduced from 2.5s to 0.4s.



NON-LINEAR LOGIC: PHASE 1



1. Model Identification

Derive the highly non-linear **Euler-Lagrange** equations for the specific multi-joint geometry.

2. Feedback Linearization

Program the controller to inject torques that exactly negate Gravity and Coriolis forces at every tick.

NON-LINEAR LOGIC: PHASE 2



3. Lyapunov Proof

Mathematically prove the proposed controller will result in $\dot{V} < 0$ for guaranteed stability.



4. Sliding Surface (SMC)

Define the s equation based on desired tracking error decay rates (exponential convergence).



Logic Architecture

Handling unmodeled disturbances via robust switching laws.

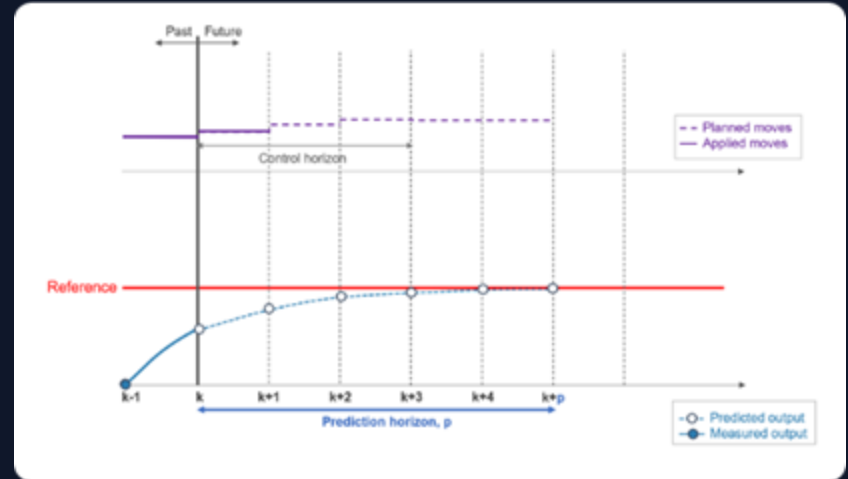
NON-LINEAR LOGIC: PHASE 3

7. MPC Horizon Setup

Choose a prediction window (e.g., 20 steps) and discretize the system for numerical optimization.

8. QP Deployment

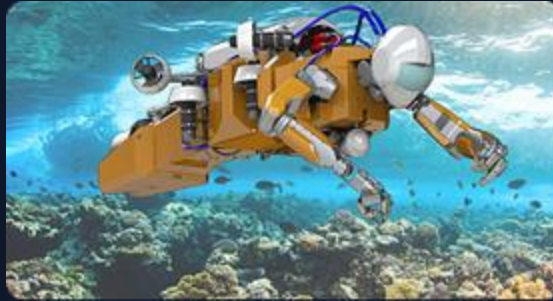
Use Quadratic Programming solvers to execute the MPC math in under 1 millisecond per cycle.



DEPLOYMENT: EXTREME AERO



Quadrotor Acrobatics: MPC executes aggressive flips while calculating thrust limits.

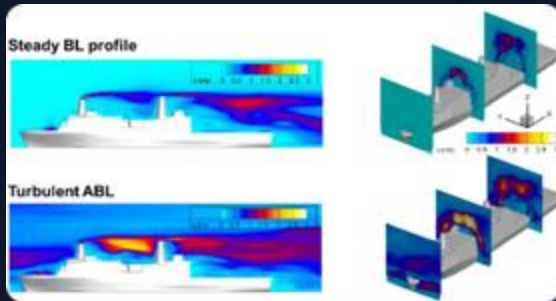


Underwater ROV: Adaptive control maintains stability against unpredictable drag currents.

 Flexible Robot

Flexible Joints: Singular Perturbation control for lightweight ultra-flexible arms.

DEPLOYMENT: ROBUST TECH



Spacecraft Reentry: SMC guarantees perfect tracking despite atmospheric turbulence.



Autonomous Drifting: MPC recalculates tire friction at 1000Hz for sideways stability.



Maglev Trains: Controlling unstable electromagnetic fields via non-linear feedback.

IMAGE SOURCES



<https://blog.robotiq.com/hs-fs/hubfs/AgWOM.png?width=496&name=AgWOM.png>

Source: blog.robotiq.com



https://apm-angidis.gr/wp-content/uploads/2022/10/%CE%A4%CE%A1%CE%9F%CE%A6%CE%9F%CE%94%CE%9F%CE%A4%CE%99%CE%9A%CE%91_Site_size.jpg

Source: apm-angidis.gr



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Source: www.youtube.com



<http://bobcad.com/wp-content/uploads/2017/02/Surface-Based-Toolpaths-Undercutting.jpg>

Source: bobcad.com



<https://image.made-in-china.com/318f0j00ZEaRVBlmbuT/Delta-Robot-8-MP4.webp>

Source: springtao.en.made-in-china.com



<https://smart-robotics.io/wp-content/uploads/2024/07/palletizing-robot.jpg>

Source: smart-robotics.io

IMAGE SOURCES



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Source: www.safetysolutions.net.au



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Source: www.mdpi.com



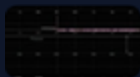
<https://www.makerforge.tech/assets/img/posts/2026-01-04-biped-leg-configuration/thumb.png>

Source: www.makerforge.tech



<https://longvista.ch/wp-content/uploads/2025/04/robotic-force-controlled-surface-polishing-68089f6fa4b36.webp>

Source: longvista.ch



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Source: www.digikey.com



<https://www.keysight.com/content/dam/keysight/en/img/cm p/use-case/frequency-response-bode-plot-measurement-use-case-banner/use-case-frequency-response-Bode-plot-Measurement-1600x900.png>

Source: www.keysight.com

IMAGE SOURCES



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Source: etimfg.com



<https://www.mathworks.com/help/mpc/gs/mpc-intro-plot.png>

Source: www.mathworks.com



<https://i.ytimg.com/vi/MvRTALJp8DM/maxresdefault.jpg>

Source: www.youtube.com



<https://www.kaust.edu.sa/PublishingImages/news/robot4.jpg>

Source: www.kaust.edu.sa



<https://builtin.com/sites/www.builtin.com/files/2024-09/soft-robotics.jpg>

Source: builtin.com



<https://www.craft-tech.com/files/urms-contours.png>

Source: www.craft-tech.com

IMAGE SOURCES



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Source: www.mdpi.com

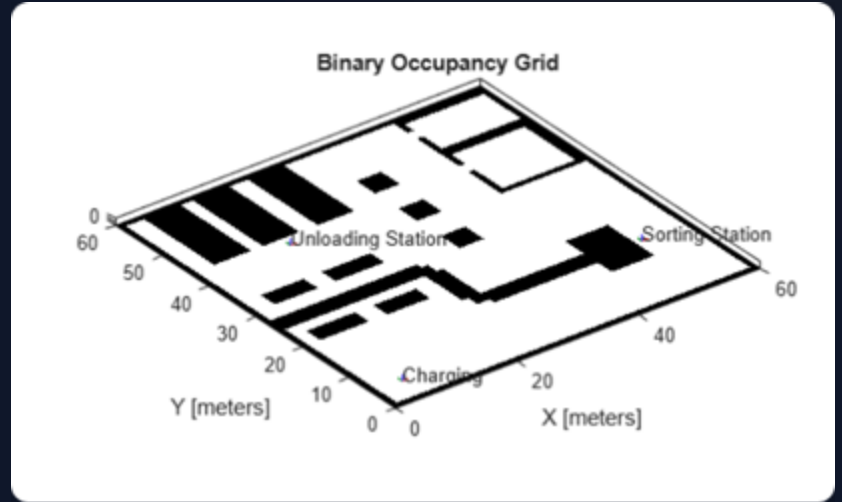
NAV LOGIC: C-SPACE & MAPPING

1. Configuration Space Mapping

Physical obstacles are "inflated" in software by the robot's specific radius. This allows treating the robot as a zero-dimension point.

2. Discretization

The continuous world is converted into a 2D occupancy grid where binary states (0=Free, 1=Wall) define the traversable graph.



Autonomous Warehouse Rover: Data Flow

Covers a complete planning example requested in Member 5.

Perception

LiDAR builds occupancy grid.
Example: 100×100 cells = 10,000 states.

Planning

A* generates route through free cells.
RRT handles crowded dynamic zones.

Execution

Trajectory smoother sends velocity commands.
Replan if obstacle appears.

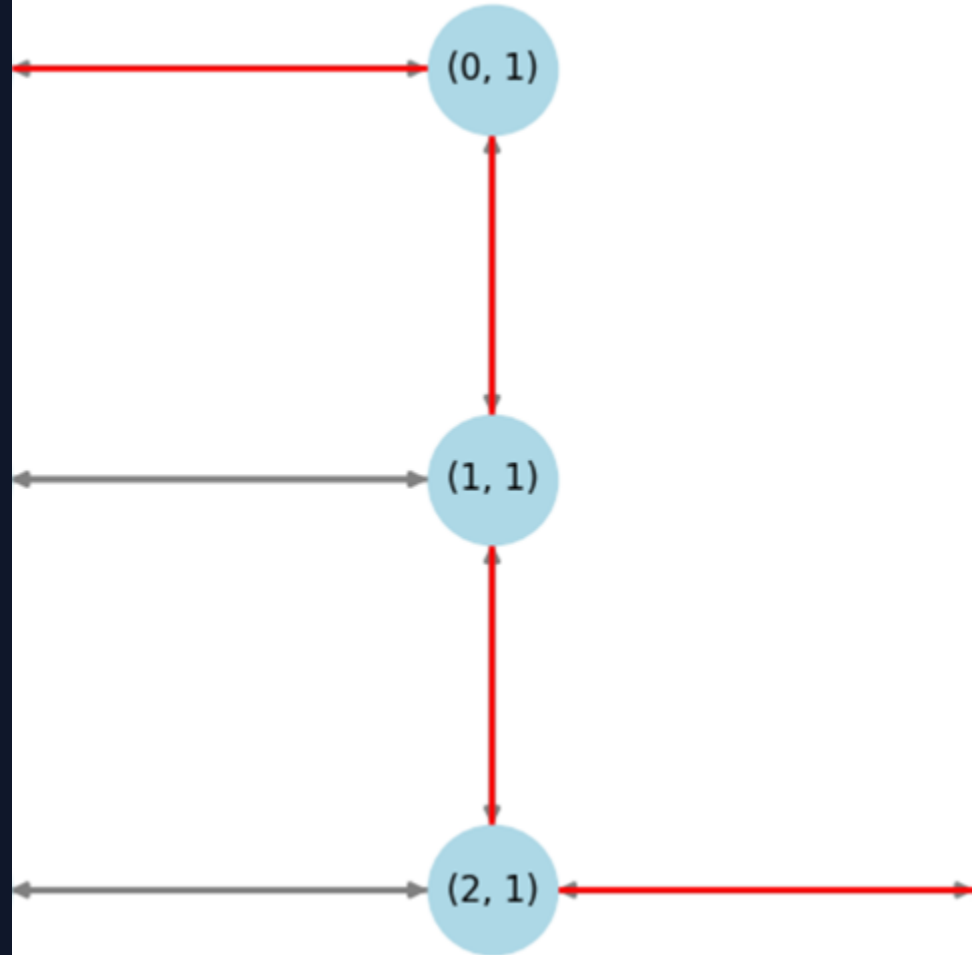
NAV LOGIC: A* SEARCH EXPANSION

3. Optimal Node Expansion

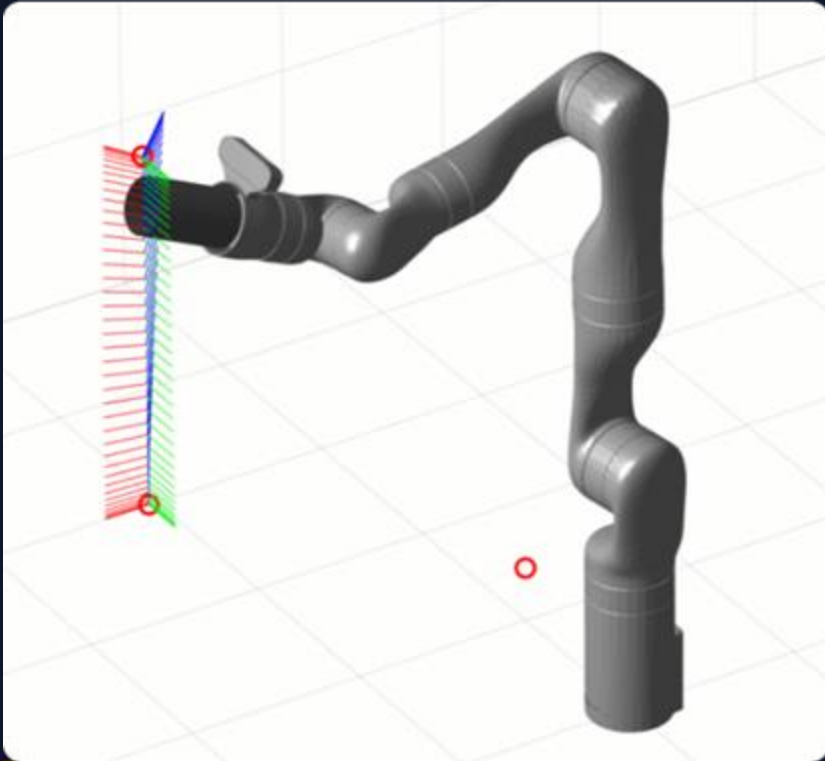
The algorithm calculates the combined cost $f(n) = g(n) + h(n)$ for all adjacent grid cells.

- ✚ Frontier management via priority queues.
- 📍 Lowest-cost node is explored first.
- 🚩 Optimal path extracted via parent back-tracing.

Graph Visualization with A* Path Highlighted



NAV LOGIC: TRAJECTORY SMOOTHING



5. Path Extraction & Smoothing

Raw algorithm paths are jagged. Software fits **Quintic Splines** through geometric waypoints.

- ✓ Continuous 1st & 2nd derivatives.
- ✓ Eliminates infinite acceleration points.
- ✓ Verifies motor torque feasibility.

DEPLOYMENT: WAREHOUSE AGVS

Amazon Kiva Logistics

Utilizing massive grid-based A* routing to coordinate thousands of mobile units.

Fleet Synergy:

Collision-free zones are negotiated in real-time, ensuring optimal throughput in high-density fulfillment centers.



DEPLOYMENT: VACUUM COVERAGE



Boustrophedon Scanning

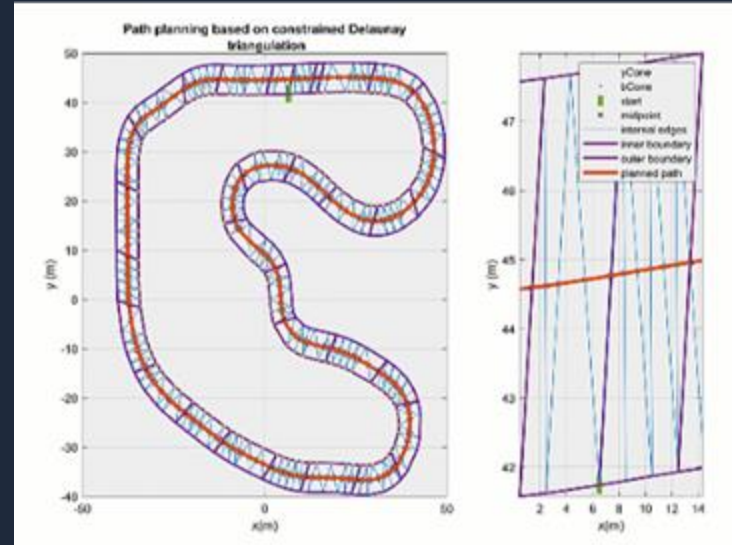
Roomba-style units use coverage path planning to guarantee every square centimeter of floor is touched.

- 🏠 Wall-following patterns.
- 🔄 Random-walk failbacks.
- 🗺️ SLAM-based internal mapping.

DEPLOYMENT: AUTONOMOUS HIGHWAY

Dynamic RRT* Planning

Planning lane changes at 70mph while predicting the trajectories of surrounding vehicles.



DEPLOYMENT: UAV NAVIGATION

Obstacle Avoidance

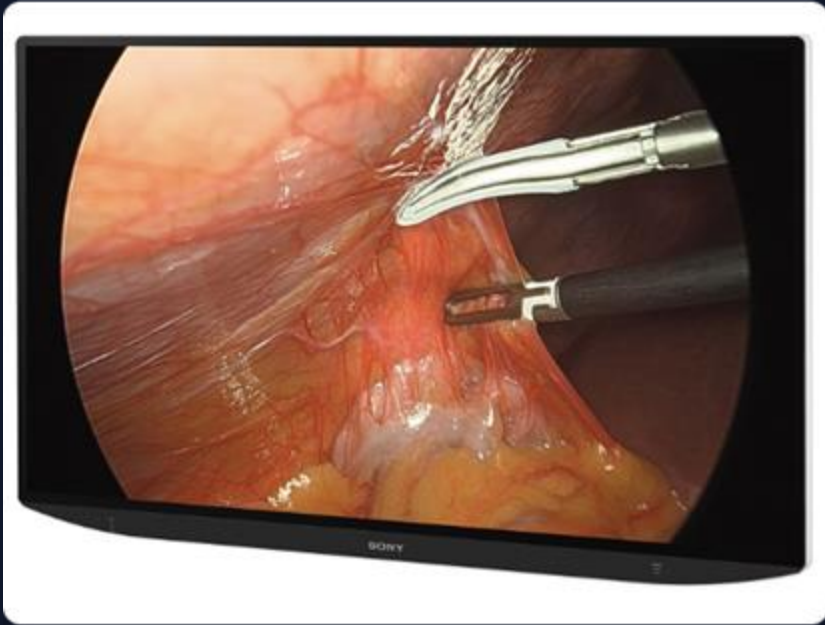
Generating 3D spline trajectories that minimize aggressive tilt angles to prevent package instability.

Reactive Shielding:

Ultrasonic and LiDAR sensors provide a 360° buffer for high-speed flight through dense foliage.



DEPLOYMENT: SURGICAL TRACKING



Vascular Pathfinding

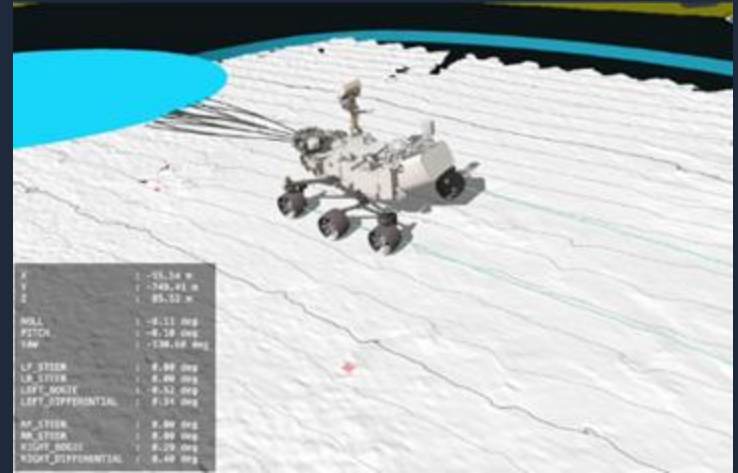
Planning flexible, snake-like paths through complex human vascular networks without tearing delicate tissue.

Sub-Millimeter Precision: Guided by real-time magnetic and visual tracking fusion.

DEPLOYMENT: MARS EXPLORATION

Local Autonomy Logic

NASA rovers execute complex pathing locally due to the 20-minute communication latency to Earth.



DEPLOYMENT: DRONE SWARMS

Multi-Agent Coordination

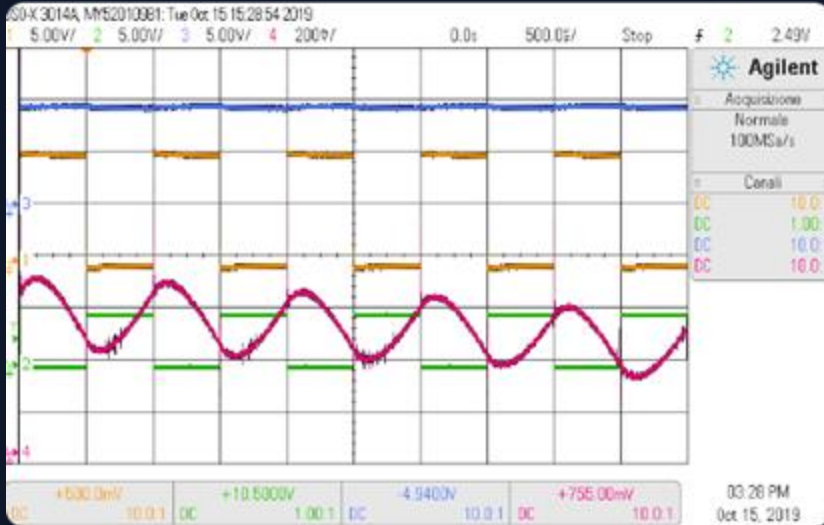
Solving 3D routing for 500+ synchronized drones during complex aerial light shows.

Anti-Collision Mesh:

Each unit maintains a dynamic potential field against its neighbors, preventing mid-air impacts.



FEEDBACK LOGIC: HARDWARE



1. Hardware Interrupts

Wiring optical encoders to dedicated hardware pins to guarantee zero missed ticks at high RPM.

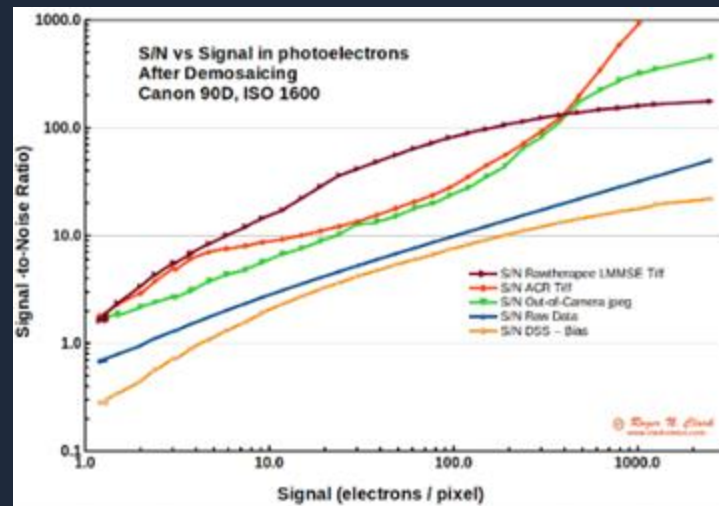
2. ADC Signal Processing

Reading raw analog voltages and applying internal scaling logic for clean sensor acquisition.

FEEDBACK LOGIC: CALIBRATION

3. Bias Removal

Calculating and subtracting zero-offset biases for IMUs placed on perfectly level surfaces.



FEEDBACK LOGIC: CONTROL LOOP



5. Filter Execution

Running Kalman matrix math at 1000Hz to output perfectly clean state variables.



6. Current Mapping

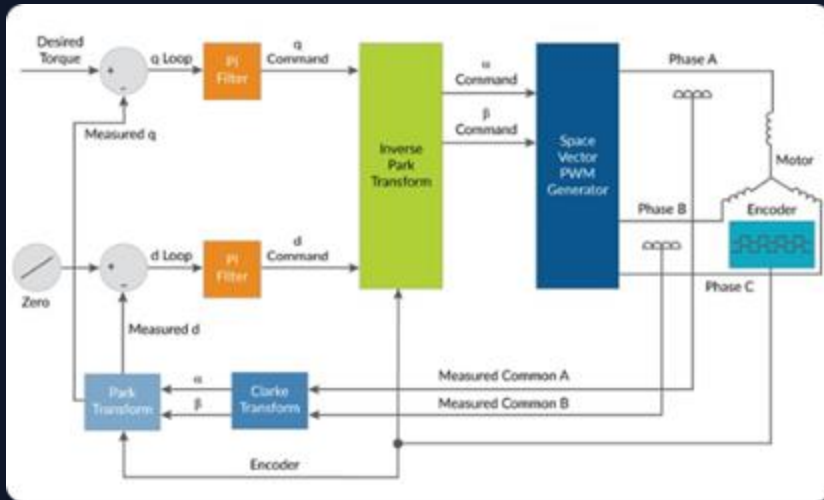
Translating desired torque into required Amperes via $I = \tau / kt$.



RTOS Sync

Maintaining deterministic cycle timing within sub-microsecond jitter margins.

FEEDBACK LOGIC: POWER DRIVE



7. BLDC Commutation

Reading Hall effect sensors to fire exact stator coils required to pull the rotor.

8. PWM Generation

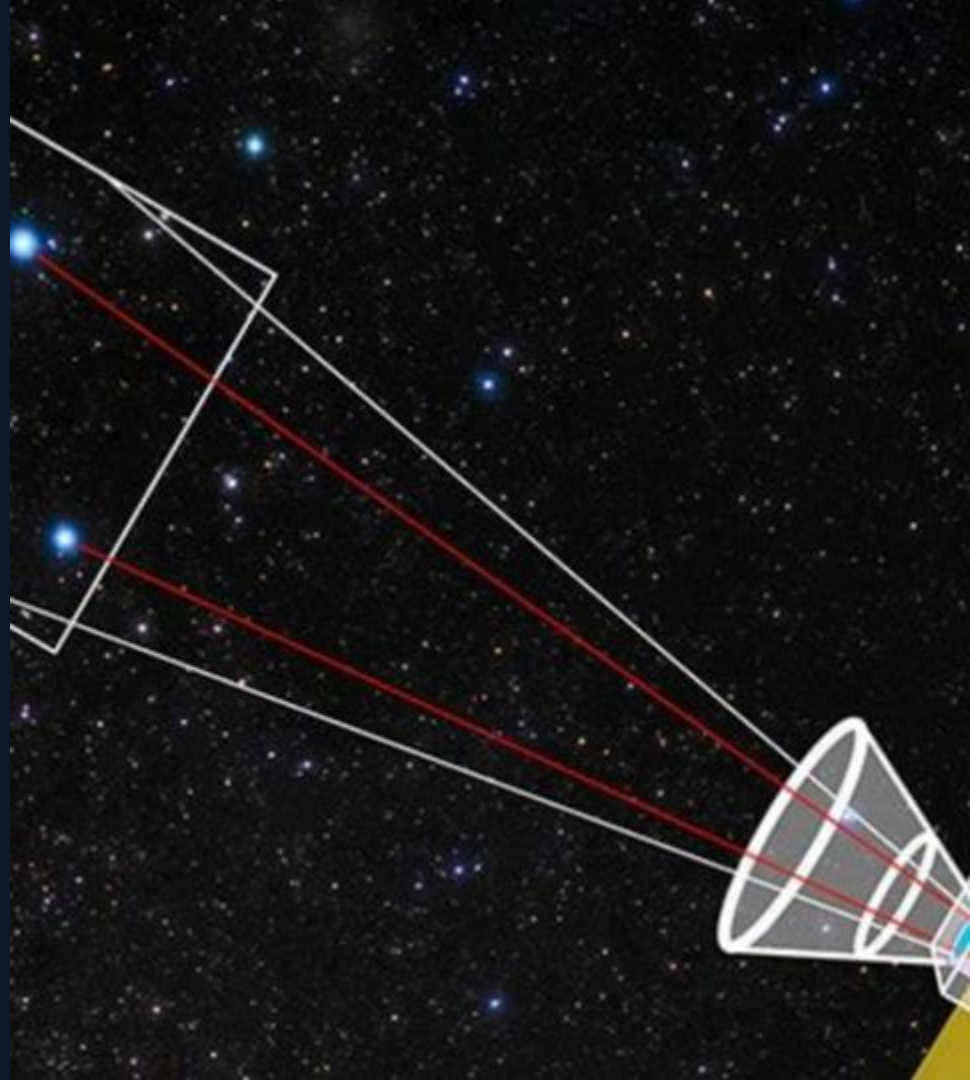
Converting desired Amperes into 16-bit PWM duty cycles for H-Bridge MOSFET gates.

DEPLOYMENT: STAR TRACKING

Spacecraft Orientation

Using **Star Trackers** fused with high-grade IMUs via Extended Kalman Filters.

Maintaining satellite pointing accuracy to within **0.001 degrees** for deep-space laser comms.



Drone Swarm Coordination

Covers multi-agent coordination as requested.

Local rule

Each drone keeps a safety bubble.
If distance < 1.5 m, repulsive action increases.
Potential-field logic prevents collision.

1.5 m

safe radius

500+

agents

sync

time control

Global rule

Central planner assigns time-stamped waypoints.
Example: 500 drones $\times$ 20 waypoints = 10,000 scheduled targets.

DEPLOYMENT: PERSONAL MOBILITY



Self-Balancing Logic

Relying on MEMS accelerometers to detect falling angles instantly, driving motors to recover the center of mass.

Inverted Pendulum Control:

High-gain PID loops reacting at 400Hz to prevent user instability.

DEPLOYMENT: INDUSTRIAL COBOTS

Active Torque Sensing

Using precise strain gauges at the joints to detect if the arm bumped a human, stopping instantly (ISO 10218-1).



DEPLOYMENT: LIDAR SLAM

Spatial Understanding

Firing millions of lasers per second to build sub-millimeter 3D point clouds of unknown rooms.

Occupancy Mapping:

Simultaneous Localization and Mapping allows the robot to track its own movement within its generated world.

DEPLOYMENT: EV POWER LOGIC



Field Oriented Control (FOC)

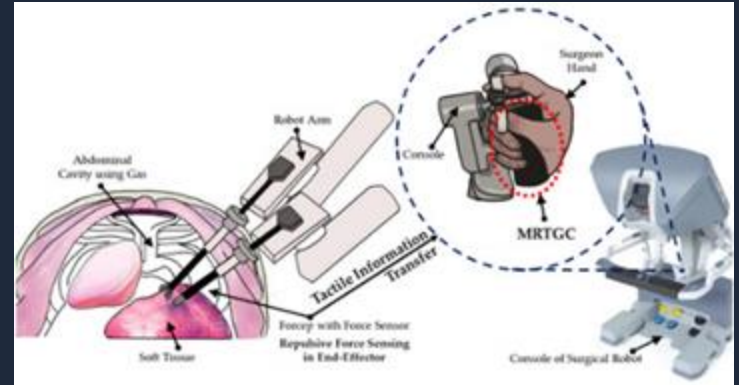
Complex Vector Control of massive 300HP AC Induction motors for smooth, silent highway acceleration.

Efficiency: Actively managing the d-q current axes to minimize reactive power losses.

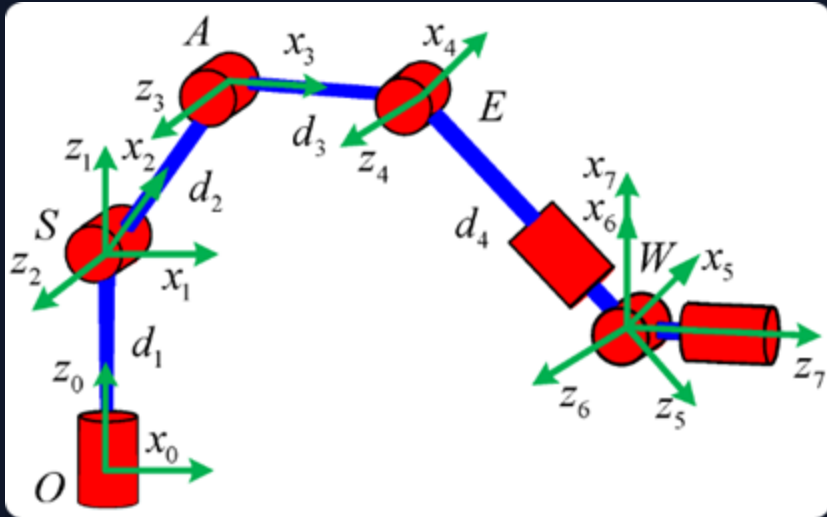
DEPLOYMENT: SURGICAL HAPTICS

Tissue Force Feedback

Translating tissue resistance measured by the robot back to haptic vibration motors on the surgeon's joystick.



ADVANCED: DIMENSIONALITY TRAP



7-DOF Search Space

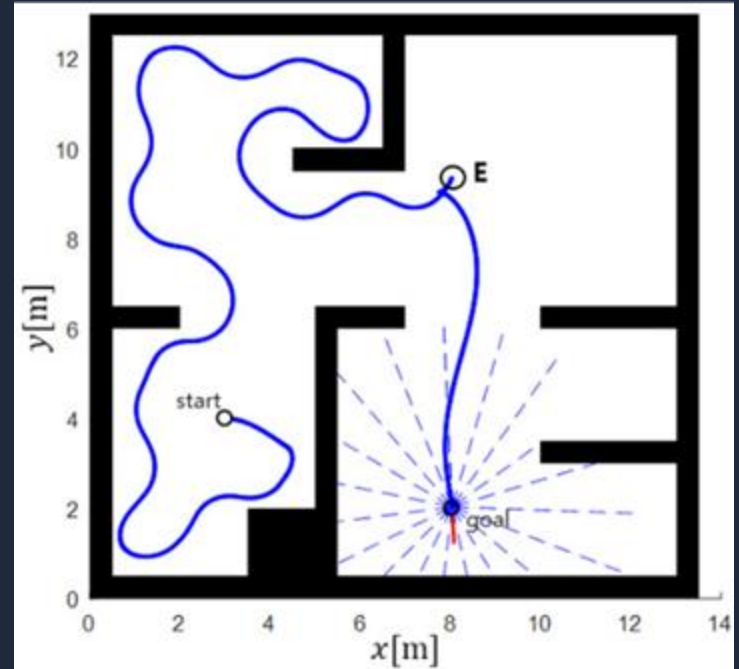
A 2D map with 100x100 pixels has 10,000 states. A 6-axis arm has **1 Trillion** states.

Grid search fails entirely at this scale, mandating randomized sampling-based methods (RRT).

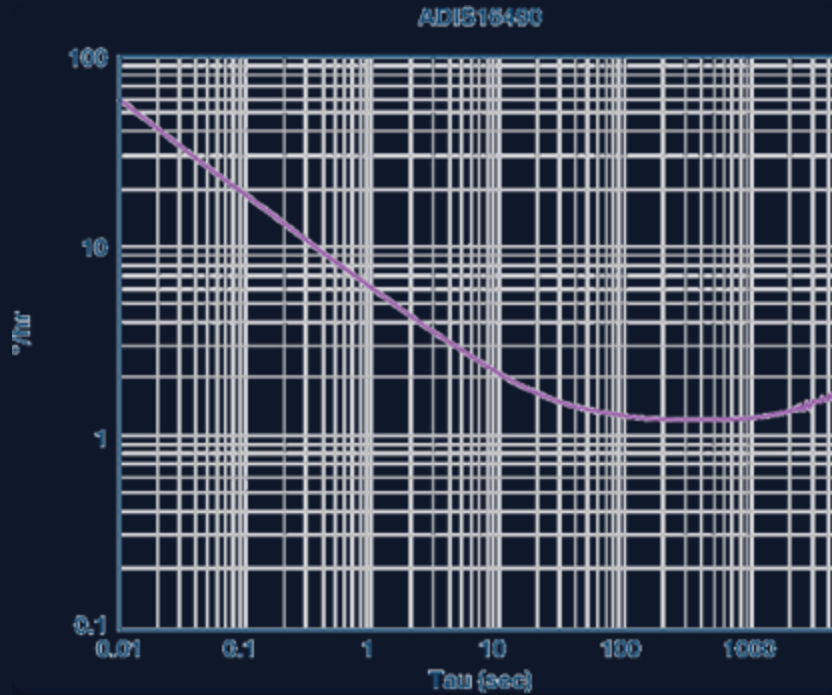
ADVANCED: LOCAL MINIMA TRAP

APF Limitations

Artificial Potential Fields are fast but notoriously trap robots in "U-shaped" corners where net force cancels to zero.



ADVANCED: SENSORY DRIFT



Integration Errors

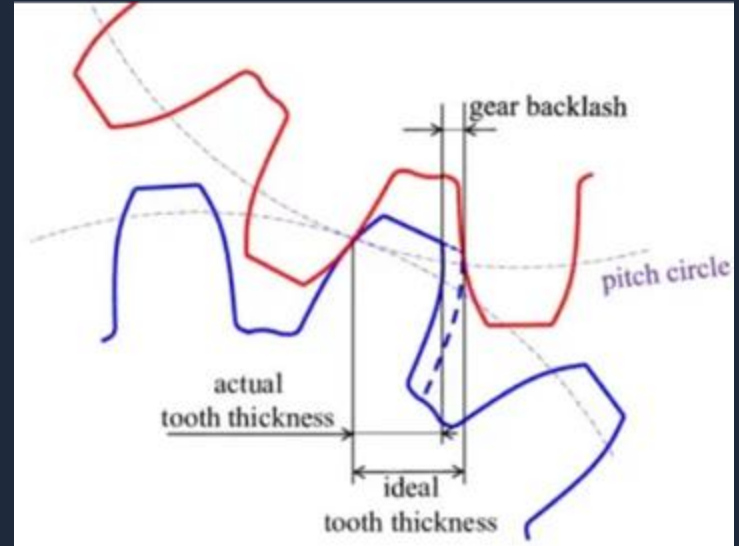
Gyroscopes measure velocity. Integrating tiny sensor noise over time results in massive accumulated heading errors.

Drones without GPS reference will inevitably crash due to this mathematical drift.

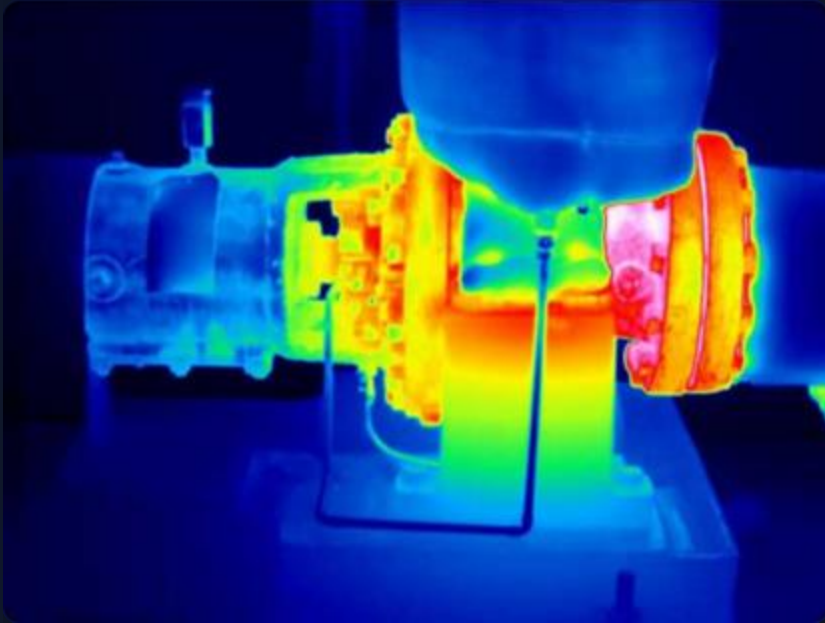
ADVANCED: GEAR BACKLASH

Mechanical Slack

When a motor reverses, it spins freely for a millimeter before the gear teeth catch. This "backlash" breaks linear assumptions.



ADVANCED: HARDWARE FAULTS



Thermal Limiters

Requesting max torque for too long burns the copper coils.
Software must calculate I^2R heat dissipation and throttle power.

Shoot-Through: Bugs in H-Bridge logic causing dead shorts,
exploding MOSFETs in milliseconds.

IMAGE SOURCES



https://ww2.mathworks.cn/help/examples/robotics/win64/ExecuteTasksForAWarehouseRobotExample_02.png

Source: ww2.mathworks.cn



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Source: www.geeksforgeeks.org



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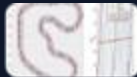
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IMAGE SOURCES



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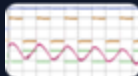
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Source: science.nasa.gov



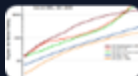
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Source: www.freepik.com



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Source: clarkvision.com

IMAGE SOURCES



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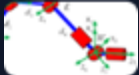
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Source: www.mdpi.com



https://www.mdpi.com/sensors/sensors-21-07257/article_deploy/html/images/sensors-21-07257-g001.png

Source: www.mdpi.com



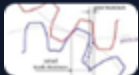
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Source: www.mdpi.com



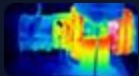
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Source: www.analog.com



<https://www.chamolgear.com/wp-content/uploads/2025/09/Backlash-Comparison-Diagram-1-edited.webp>

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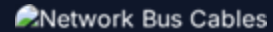
COMMUNICATION LATENCY

Network Delay Impact

If sensors communicate over slow serial buses, controllers compute commands based on past states.

Bus Type	Typical Latency
Ethernet	2.0 ms (Variable)
CAN-Bus	0.5 ms (Stable)
EtherCAT	0.05 ms (Hard Real-Time)

Delay in feedback loops is a fundamental cause of system destabilization.



Network Bus Cables

System Integration, Case Studies & Future Frontiers

Covers the grand-finale layer: ROS, RTOS safety, major robotics systems, and future trends.

ROS Architecture: Nodes, Topics, Services

Covers the requested software architecture foundation.

Nodes

Independent programs: camera node, planner node, motor node.
Each has one responsibility.

Topics

Streaming channels for data.
Example: /scan for LiDAR, /cmd_vel for velocity.

Services / actions

Request-response or long tasks.
Example: "navigate to waypoint A".

Camera node

/image

Planner node

/cmd_vel

Motor node

REDUNDANCY ARCHITECTURE



Triple Modular

Aerospace systems use 3 identical sensors. Software "votes" to ignore outliers.



Fail-Safe Modes

Determining "Safe-Stop" states in the event of primary sensor loss or power surge.



Hardware Voting

Hardware-level comparison logic ensures no single point of failure (SPOF).

RTOS + Failsafe Protocols

Adds safety and failsafe coverage for human-robot collaboration.

Hard real-time

Motor loop must execute every 1 ms.
Late command = unstable control.

Safe stop

If sensor mismatch or watchdog timeout occurs,
power is reduced and brakes engage.

Redundancy

Triple sensors vote 2-of-3.
Single bad sensor cannot crash the robot.

BRUSHLESS DOMINANCE

Why BLDC Prevails

Brushed motors create physical sparks, suffer from wear, and have strictly limited RPM caps.

- ✓ No touching parts = Long Life
- ✓ Higher Torque-to-Weight ratio
- ✓ Precise Electronic Commutation

KALMAN UNCERTAINTY LOGIC

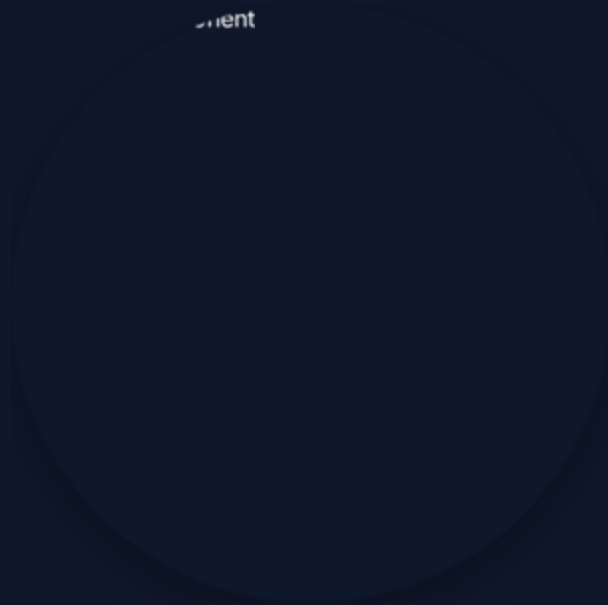
Covariance Matrix (P) Evolution

Uncertainty Growth (Blind Move)



Kalman doesn't just filter; it tracks the **Uncertainty** of its own guesses dynamically.

H-BRIDGE SHOOT-THROUGH



Dead-Short Risk

If software bugs cause the top and bottom MOSFETs of an H-bridge to fire simultaneously, it creates a dead short.

Result: Instant thermal explosion of the microchip.

Solution: Hardware-level interlocking circuits and dead-time insertion.

JITTER VS. DETERMINISM

Real-Time Operating Systems

Robotics requires RTOS. If a sensor is read at 1000Hz, but Linux interrupts it for 2ms to check Wi-Fi, the discrete math collapses.

$$\Delta t = t_{\text{nominal}} \pm \text{jitter}$$

High jitter creates non-deterministic gains, leading to high-frequency instability.

 Oscilloscope Jitter

ENCODER TECHNOLOGY



Optical Encoders

Immune to motor magnets but vulnerable to dust and environmental debris.



Magnetic Encoders

Survives dirty environments but suffers from non-linear magnetic interference from coils.



DIRECT DRIVE REVOLUTION

The Death of Gears

Modern high-end robots are ditching gearboxes entirely to avoid backlash hysteresis.

Mechanics: Massive, custom-wound motors directly connected to the legs.

Result: Near-infinite back-drivability and ultimate force sensitivity.

CONCLUSION

Mastering Physics

A master of Kinematics and Dynamics commands the physical reality of the robot perfectly.

Code cannot control physics directly. Sensors and Actuators are the noisy translation layers that make Robotics an elite engineering discipline.

Case Study: Boston Dynamics-Style Legged Robot

Covers one major system breakdown example.

Perception layer

Cameras + depth sensing estimate terrain height and obstacles.

Control layer

Whole-body MPC computes ground reaction forces for each foot.

Actuation layer

High-bandwidth joints track torque commands while absorbing impacts.

balance

CoM control

contact

force planning

terrain

adaptive gait

Case Study: da Vinci Surgical System

Adds the medical robotics system requested in the outline.

Master console

Surgeon joystick commands are scaled and filtered to remove hand tremor.

Slave arms

Precise cable-driven robot tools execute tiny movements inside the patient.

Safety logic

Motion limits, clutching, visual feedback, and redundant sensing prevent unsafe commands.

Case Study: Autonomous Vehicle Control Stack

Adds the autonomous vehicle grand-system example.

Perception

Cameras, radar, LiDAR detect lanes, cars, pedestrians, and traffic signs.

Prediction + planning

System predicts other agents and selects safe trajectories.

Control

Steering, throttle, and braking track the chosen trajectory with feedback loops.

Sensors

Perception

Prediction

Planning

Control

ACADEMIC DATABASE

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Navigation bridges the gap between pure mathematics and spatial reality. A brilliant controller is useless if the planned path hits a wall.

— Ahmad Safi, Trajectory Lead



TEAM RETROSPECTIVE



Collaboration

Integrating 5 core disciplines into a unified state-space architecture.



Validation

Thousands of hours of simulator and hardware-in-the-loop testing.



Innovation

Pushing the boundaries of autonomous control at JUST.

The Next Frontier

Moving toward End-to-End Deep Reinforcement Learning for multi-agent coordinated humanoid dynamics.



Future Trend: Soft Robotics

Covers one future-frontier topic requested in the instructions.

Why it matters

Soft robots use flexible materials instead of rigid links.
They can safely interact with people, food, tissue, and fragile objects.

Control challenge

Shape is continuous, not just joint angles.
Models require pressure, deformation, and material stiffness.

safe

human contact

nonlinear

material model

medical

high-value use

Future Trend: Brain–Computer Interfaces for Prosthetics

Adds the second future frontier requested.

Signal path

Neural or muscle signals are decoded into intended movement.
Example: “close hand” → prosthetic grip command.

Robotics link

Control system maps intention to actuators while sensors return grip force.
Closed-loop feedback makes the prosthetic feel more natural.

Future Trend: Quantum Control Concepts

Adds the final future-frontier topic while keeping it academically careful.

What it means here

Quantum control is not mainstream robot control today.
It is a research frontier for controlling quantum states, sensors, and future precision systems.

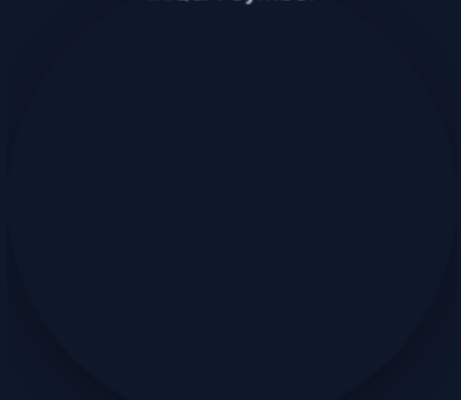
Robotics relevance

Possible future link: ultra-precise quantum sensors for navigation where GPS fails.
Current use is mostly theoretical / research-oriented.

Questions?

Interactive discussion and system demonstration.

 Q&A Symbol



How the Full Deck Connects

A final map connecting the six member areas into one robotics control story.

Member 1–2

PUMA 560
Rigid transforms
Inertia tensor
Euler–Lagrange
Newton–Euler
Inverted pendulum

34+

integrated topics

Member 3–4

Root locus
Bode margins
Hydraulics/resolvers
State-space
LQR
Impedance/admittance

6/6

members covered

Member 5–6

SLAM pipeline
AI vision
Parallel compute
ROS
Major case studies
Future trends

same theme

preserved

TRANSMISSION HALTED

End of Study

Jordan University of Science and Technology

Research Team Alpha | Amman, Jordan

IMAGE SOURCES

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